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Highlights

- We compare high-frequency measures of liquidity with easy to compute low-frequency measures for the two cryptocurrencies bitcoin and ethereum.
- Low frequency liquidity measures are relatively good estimates of actual liquidity in cryptocurrency markets.
- Spread estimators based on high and low prices capture the time-series variation of actual bid-ask spreads well.
- The Amihud (2002) illiquidity ratio and the Kyle and Obizhaeva (2016) estimator perform best when estimating the level of the liquidity benchmark measures.

How to Measure the Liquidity of Cryptocurrency Markets?[☆]

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Abstract

This paper investigates the efficacy of low-frequency transaction based liquidity measures to describe actual (high-frequency) liquidity. We show that the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) estimators outperform other measures in describing time-series variations, irrespective of the observation frequency, trading venue, high-frequency liquidity benchmark, and cryptocurrency. Both measures perform well during high and low return, volatility and volume periods. The Kyle and Obizhaeva (2016) estimator and the Amihud (2002) illiquidity ratio outperform when estimating liquidity levels. These two estimators also reliably identify liquidity differences between trading venues. Overall, the results suggest that there is not yet a universally *best* measure but there are reasonably *good* low-frequency measures.

Keywords: cryptocurrencies, liquidity, capital markets

JEL: G12, G14

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1. Introduction

Bitcoin and other cryptocurrencies are now firmly entrenched in the financial system. Bitcoin is becoming a widely accepted form of online payment and more than 35 million bitcoin wallets are in existence. Trading in bitcoin exceeded \$930 billion USD in January 2020. Bitcoin also forms a growing part of investment portfolios and serves as the underlying for futures contracts,¹ recently exceeding \$1 billion in open interest.² Originally designed as a decentralized digital cash system using cryptographic hash functions to secure transactions, it is poised to overtake national fiat currencies and other financial assets in terms of global importance.

Bitcoin and other cryptocurrencies are traded on numerous trading platforms around the globe. Bitcoin can be traded 24 hours per day, and seven days per week for US dollars, the Euro, Japanese Yen, as well as numerous other fiat and crypto(currencies). Monthly dollar trading volume on the New York Stock Exchange was of similar magnitude to global bitcoin trading, with \$1.03 trillion USD for the NYSE during December 2020.³

The growing importance of bitcoin for payments and investments is dependent on an efficient transfer of bitcoin for other currencies on cryptocurrency exchanges. The number of exchanges has exploded, making it difficult for investors to select an exchange for trading and hedging. While trading has become relatively frequent in cryptocurrencies the liquidity of these markets is difficult

¹See: <https://www.cmegroup.com/trading/bitcoin-futures.html>

²<https://www.coindesk.com/bitcoin-futures-pass-1b-in-open-interest-on-bitmex-for-first-time-since-march-crash>

³Data for bitcoin volume come from <https://coinmarketcap.com/currencies/bitcoin/historical-data/> and NYSE volume data are from <https://focus.world-exchanges.org/issue/january-2020/market-statistics>. Both sites were accessed on February 15, 2020. We note, though, that some cryptocurrency exchanges appear to be overstating their reported volume (e.g. Hougan et al. (2019)) and that ranking websites like coinmarketcap.com may be tempted to report inflated trading volume due to their revenue model which is largely dependent on crypto-exchanges (Alexander and Dakos (2019)).

to determine. Cryptocurrency markets lack a regulated data feed like the consolidated tape for U.S. equities. The lack of a consolidated feed, coupled with the high number of exchanges and jurisdictions makes it difficult to calculate high-frequency bid-ask spreads thereby hampering the comparison of liquidity across cryptocurrency exchanges. The bid-ask spread is an important metric when assessing an exchange in that it represents the costs of immediately buying or selling a security. Bid-ask spreads are usually calculated using high-frequency intraday data that are both expensive to purchase and time-consuming to process. We compare high-frequency measures of liquidity with easy to compute low-frequency measures.

Characterising liquidity across exchanges is important for investors, traders, and hedging strategies that use cryptocurrencies (Hu et al. (2019)) that can be negatively affected by the costs of illiquidity. Additionally, cryptocurrency prices are not integrated across exchanges (Makarov and Schoar (2020)) and the decision to trade on an exchange is binding as orders cannot easily be re-routed to exchanges that are more liquid or offering better prices. With little information about individual exchanges traders may have to rely on transactions data such as the daily high, low and closing prices to evaluate market quality.

We study the accuracy of liquidity measures derived from transactions data. To this end we estimate low-frequency measures derived from aggregate transactions data⁴ (prices and volumes) and compare them to high-frequency measures of transaction costs and price impact calculated from order book data. Our objective is to identify the transactions-based measure that best describes actual liquidity on a cryptocurrency exchange.

Data on best bid and ask prices and order books are hard to obtain and

⁴Below, we use the terms transactions-based measure synonymously with the terms low-frequency measure and liquidity proxy.

process.⁵ As such, few papers use full order book data to study the liquidity of cryptocurrency markets (Brauneis et al. (2019), Dyhrberg et al. (2018), Hautsch et al. (2018), Makarov and Schoar (2020) and Marshall et al. (2019)).

50 A number of low-frequency measures have been developed and used to assess bond, commodity, foreign exchange and equity market liquidity (e.g. Fong et al. (2017), Goyenko et al. (2009), Marshall et al. (2012), Karnaukh et al. (2015), Schestag et al. (2016), Johann and Theissen (2020)). Cryptocurrency markets have characteristics that differ from traditional markets,⁶ suggesting that liq-

55 uidity formation on cryptocurrency exchanges may differ from those of other asset markets.

We use a novel and comprehensive set of continuous transactions data and order book snapshots comprising the 50 best bids and asks for two major cryptocurrencies (bitcoin and ethereum) and three large exchanges (Bitfinex, Bitstamp and Coinbase Pro) over a two-year period. First, we use these data to construct high-frequency measures of transaction costs and price impact. These measures serve as our liquidity benchmarks. In a second step, we use transactions data (prices and volumes) and calculate various liquidity proxies at lower frequencies (1 hour, 1 day, and 15 days,⁷ respectively). Data to compute the

60 measures are collected at the 1-minute, 1-hour and 1-day frequency.⁸ Individual low-frequency measures have been used to describe liquidity in cryptocurrency markets (e.g. Brauneis and Mestel (2018), Dimpfl (2017), Fink and Johann

⁵It can be downloaded in real time from the REST APIs of each cryptocurrency exchanges or it can be purchased from vendors such as Kaiko.

⁶e.g. markets are highly fragmented and weakly regulated; 365 days a year and a 24 hour trading a day, and direct market access for all traders; trading platforms allow a direct transfer of fiat currency from and to bank accounts or credit cards and transactions are cleared and settled by exchanges directly; margin trading and short-selling is uncommon

⁷Providing results at the monthly frequency is infeasible because high-frequency data for cryptocurrencies are limited to 24 months.

⁸In contrast to CRSP for equity markets that provides daily prices, cryptocurrency prices are available at higher than daily frequency. For example, the site cryptodatadownload.com provides free data for many currency pairs and trading venues at the hourly frequency.

(2014) and Shi (2018)) but the relative benefits of each is not well understood.

The most commonly used of these low-frequency measures are the Roll (1984) serial covariance estimator and the Amihud (2002) illiquidity ratio. We extend the
 70 analysis to include low-frequency measures based on high and low prices (Abdi and Ranaldo (2017), Corwin and Schultz (2012)), the volatility-to-volume measure proposed by Kyle and Obizhaeva (2016) and simple volume-based measures (the transaction frequency and the USD trading volume). Our goal is to evalu-
 75 ate these measures' ability to describe time-series variation in liquidity as well as level differences across exchanges. This will allow us to recommend a specific liquidity proxy using easy-to-access transactions data and an easy-to-compute proxy. Using such a proxy offers enormous savings compared to high-frequency order book measures.

80 Our paper augments the literature on low-frequency transactions-based liquidity measures by extending the analysis to cryptocurrencies, an important and emerging asset class. We focus on results for bitcoin, the largest cryptocurrency. Results for ethereum, which are qualitatively similar in most respects, are discussed in section 3.7, the corresponding tables and figures are in the appendix.
 85 We use the cost of a roundtrip trade in addition to quoted spreads, effective spreads and price impacts as benchmark measures. The round-trip measure provides estimates of the execution costs for large trades and is thus important for evaluating some trading strategies (Hu et al. (2019)) and factor models (Liu et al. (2019)).

90 The results suggest that the proxies that use high, low and closing prices, the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) estimators, best capture the time-series variation in cryptocurrency liquidity. These measures work for all data frequencies, exchanges (Bitfinex, Bitstamp, Coinbase Pro), benchmark measures (quoted spread, effective spread, price impact, cost of a roundtrip

trade) and for both bitcoin and ethereum.⁹ Average time-series correlations describe the average relationship between benchmark measures and proxies but do not capture the relationship for extreme liquidity events that may be important for investment and hedging strategies. We use quantile dependence plots to understand how well transactions-based liquidity measures capture the time-series properties of the benchmark measures across the distribution. This is an important extension since the relative performance of liquidity proxies might be different depending on the liquidity regime. We perform several sample splits and find similar performance rankings of our liquidity proxies for high and low volume and volatility periods. Given the extreme volatility associated with bitcoin and cryptocurrency markets more generally, identifying liquidity proxies that are invariant to volatility and volume swings is an important contribution.

The popular Amihud (2002) illiquidity ratio does not capture the time-series variability of liquidity in the cryptocurrency markets.¹⁰ The poor performance is driven by the relationship between volume and liquidity that is assumed to be negative in Amihud (2002) and is positive in cryptocurrency markets. The positive relation between bid-ask spreads and volume is at odds with most theoretical predictions but has recently also been documented by Bogousslavsky and Collin-Dufresne (2020) for large US stocks.

Consistent with prior efforts to identify good low-frequency proxies of liquidity (Hasbrouck and Seppi, 2001), we construct a composite estimator, the first principal component of the low-frequency proxies, and find that it does not improve on the performance of the best individual proxies using intraday data, but offers some improvements at the daily and lower frequencies.

⁹Consistent with our results, Karnaukh et al. (2015) report the Corwin and Schultz (2012) measure to have the highest correlation with high-frequency bid-ask spreads in FX markets.

¹⁰Conceptually, the Amihud (2002) illiquidity ratio is a proxy for the price impact, not for the spread. We find that it also performs poorly in tracking the time-series variation of price impacts, a component of the effective spread.

The measures that best describe the level of the benchmark measures are the
 120 Kyle and Obizhaeva (2016) and Amihud (2002) estimators. In this application
 the Corwin and Schultz (2012) and Abdi and Rinaldo (2017) estimators perform
 poorly. We find that the values obtained for these two estimators and for the Roll
 (1984) estimator are negatively related to the data frequency, a finding that has
 been documented previously for the Roll estimator (Roll (1984), Harris (1990))
 125 but has, to the best of our knowledge, not been documented for the high-low
 spread estimators.

An important application of liquidity proxies is to select an execution venue
 among a number of alternatives. We use the low-frequency estimators to rank
 trading venues according to their liquidity. We find that the Amihud (2002)
 130 illiquidity ratio and the Kyle and Obizhaeva (2016) estimator replicate the 'true'
 ranking when compared to the ranking generated using high-frequency order
 book measures.¹¹

Our findings are useful for researchers, investors, traders, trading venue op-
 erators and regulators to understand liquidity levels and dynamics on cryptocur-
 135 rency exchanges with relatively easy to acquire and process aggregate price and
 volume data. Investors seeking the most liquid exchanges are best advised to use
 the Amihud (2002) illiquidity ratio or the Kyle and Obizhaeva (2016) estimator.
 These two measures also provide good approximations of the level of liquidity.
 They are the measures of choice for market participants attempting to estimate
 140 execution costs to evaluate trading strategies. In contrast, traders seeking to
 time the liquidity of cryptocurrency markets and enter or exit when markets are
 liquid should use the Abdi and Rinaldo (2017) or Corwin and Schultz (2012)
 estimator as they best capture the time-series variability of the quoted and effec-

¹¹The Corwin and Schultz (2012) estimator does very well for bitcoin but does poorly for
 ethereum.

tive bid-ask spread. Regulators and trading venue operators can learn from our
 paper about how exchanges compare across time and use aggregate measures to
 study the impact on liquidity of regulatory or market changes. Researchers can
 use our results to guide their choice of liquidity measures in empirical studies
 on cryptocurrency markets. Overall, our results suggest that the measure used
 should depend on the question being asked, as there is not (yet) a universally
 best measure.

The remainder of the paper is organized as follows. In section 2 we de-
 scribe our data and methodology, section 3 presents the results, and section 4
 concludes.

2. Data and Methodology

2.1. Data

We compile a high-frequency data set that covers the two years period from
 12/16/2017 00:00 UTC to 12/16/2019 00:00 UTC, a total of 730 trading days
 (17,520 hours). Over this period we used Matlab to continuously access the
 public and freely accessible REST APIs of three large trading venues, Bitfinex,
 Bitstamp and Coinbase Pro (formerly known as GDAX). These are among the
 largest cryptocurrency spot trading platforms. All three venues operate an
 electronic central limit order book with orders being matched based on price
 and time priority.

The REST APIs provide live information on transactions and the current
 state of the order book. All public endpoints at each of these exchanges use
 GET requests for different types of information. We request records on 'Trades'
 / 'Transactions' and the 'Order book'. Depending on the venue, request parame-
 ters vary. For instance, Bitstamp only provides the full order book (with usually
 thousands of entries) whereas order book requests at Bitfinex and Coinbase Pro

170 may be limited to the 50 best orders on each side of the market.

A potential problem associated with transactions data from cryptocurrency exchanges are fake data. A widely cited report by Hougan et al. (2019) argues that up to 95% of exchange-reported trading volume in bitcoin might not represent economically meaningful transactions or might even be plain fake. Collecting unique high-frequency trade and order book data for bitcoin the authors subject 83 cryptocurrency exchanges to several tests to identify exchanges that are likely to overstate trading volume. Only 10 exchanges passed all the tests and are characterized as "real volume" exchanges. The three trading venues that we consider in the present study all belong to the latter group.

180 From each trading venue we download data for two currency pairs, bitcoin versus US dollar (BTCUSD) and ethereum against US dollar (ETHUSD). The data set includes the price and the corresponding dollar trading volume for each transaction, a UNIX time stamp, a unique exchange-specific ID and a trade indicator which indicates whether a transaction was buyer-initiated or seller-initiated. Table 1 lists the total number of transactions and order book snapshots for both currencies and all three markets. A total of 90.7 (53.4) million transactions were executed for bitcoin (ethereum) during the investigation period, most of them on Coinbase Pro while Bitstamp reports least transactions.

INSERT TABLE 1 ABOUT HERE

190 We observe several time intervals with gaps in the data. These may be due to actually missing trading activity, technical problems (failure of the internet connection, no response from the server etc.), or exchange-specific trading halts (e.g. due to maintenance, updates or hacker attacks). We identify between 6,329 (Coinbase Pro - BTC) and 187,254 (Bitstamp - ETH) intervals without transaction data exceeding 60 seconds (1 minute), between 2,641 and 5,920 intervals exceeding 600 seconds (10 minutes) and between 1,573 and 2,199

intervals exceeding 1,800 seconds (30 minutes).

Table 2 provides descriptive statistics. Trading activity is markedly higher for BTCUSD than for ETHUSD on all three exchanges. The differences are more pronounced for the USD trading volume than for the number of transactions, implying that the average trade size is smaller for the currency pair ETHUSD. With respect to the number of transactions Coinbase Pro (Bitstamp) is the most active (least active) exchange for both currencies. However, average daily USD volume is highest on Bitfinex and lowest on Bitstamp. Concerning average USD trade size Bitstamp (Coinbase Pro) has the highest (lowest) level for the pair BTCUSD, while for ETHUSD Bitfinex (Coinbase Pro) shows the highest (lowest) level. The standard deviation of ETHUSD returns is larger than that of BTCUSD returns on all three venues. Across venues, returns are most volatile on Bitstamp and least volatile on Coinbase Pro.¹²

INSERT TABLE 2 ABOUT HERE

Besides transactions data we retrieve order book data from the three trading platforms. Specifically, we collect the 50 best bid and best ask prices with corresponding volumes, resulting in a total 14.6 million (13.5 million, 16.4 million) order book snapshots for Bitfinex (Bitstamp, Coinbase Pro) for the two cryptocurrencies under investigation (see Table 1). As for the transactions data we observe a considerable number of intervals without order book snapshots. There are between 13,038 (Coinbase Pro) and 84,461 (Bitfinex) intervals without data exceeding 60 seconds. The numbers of intervals without order book snapshots exceeding 600 seconds and 1,800 seconds are roughly equal across the three exchanges and amount to approximately 3,300 and 2,200, respectively. The

¹²We note that the differences in the standard deviation of returns may reflect liquidity differences because the return standard deviation is affected by bid-ask bounce. In fact, as shown in Table 3 below, bid-ask spreads are largest on Bitstamp, a result that has also been confirmed by Brauneis et al. (2019).

standard deviations of quote midpoint returns is similar across trading venues and is generally higher for ETHUSD than for BTCUSD (see Table 2).

2.2. Measures of liquidity

The purpose of our paper is to assess and compare the accuracy of transactions-based measures of liquidity. In doing so we take the perspective of a researcher who has access to data on open, high, low and close prices and on the number of transactions and the dollar trading volume.

For our analysis we need to specify (a) the frequency at which these data are available (measured by the length of the *subintervals* i in the sequel) and (b) the frequency at which the transactions-based measures are calculated (measured by the length of the *intervals* t). Unlike for other financial markets (e.g. stock markets), price and volume data for cryptocurrencies are easily available for higher than daily frequencies. We therefore choose three distinct setups.

- Data are available at the 1-minute frequency and are used to estimate transactions-based liquidity measures at the hourly frequency.
- Data are available at the 1-hour frequency and are used to estimate liquidity measures at the daily frequency.
- Data are available at the daily level and are used to calculate liquidity measures at a 15-day frequency.

To construct the data set we use our record of all transactions and extract the open, high, low and close price as well as the number of transactions and the dollar volume at the respective frequencies of one minute, one hour and one day. These data are then used to calculate the transactions-based measures (to be described below) at the hourly, daily and 15-day frequencies, respectively. Because the three trading venues under investigation are located in different

time zones we follow `coinmarketcap.com` and define a trading day as lasting from 00:00 UTC to 23:59 UTC. For an interval to be included in the analysis we require that data are available for at least 80% of the subintervals. Thus, when we aggregate minute-by-minute (hour-by-hour, daily) data to the hourly (daily, 15-day) frequency we require at least 48 minutes (19 hours, 12 days) with valid data. The final data set roughly comprises 12,500 hourly intervals, 670 daily intervals and 46 15-day intervals respectively.

Besides the transactions-based measures we need to calculate benchmark measures. To this end we use the complete record of all transactions and all order book snapshots and calculate average quoted and effective spreads, price impacts and cost of a roundtrip trade (to be defined below) at the hourly, daily and 15-day frequency.

In the sequel we first describe the high-frequency measures which we use as benchmark measures and then the transactions-based measures that we wish to evaluate.

2.2.1. High-frequency benchmark measures

- Percentage Quoted Spread (QS)

The percentage quoted spread is the difference between the best ask price P^a and the best bid price P^b of each order book snapshot, divided by the quote midpoint $MQ = (P^b + P^a)/2$ and averaged over all observations in the interval

$$QS_t = \frac{1}{N_t} \sum_{j=1}^{N_t} \frac{P_j^a - P_j^b}{MQ_j}.$$

The subscript j denotes the j^{th} order book snapshot in interval t and N_t is the total number of order book snapshots in interval t .

- Percentage Effective Spread (ES)

To estimate the effective bid-ask spread we combine order book snapshots

with the first transaction that occurs after the snapshot.¹³ The price of this transaction is denoted P^+ . The average percentage effective spread in interval t is then calculated as

$$ES_t = \frac{1}{N_t^+} \sum_{j=1}^{N_t^+} \frac{2 \cdot |P_j^+ - MQ_j|}{MQ_j}$$

N_t^+ is the number of order book snapshots that are followed by a transaction before the next order book snapshot is recorded.

- Percentage Price Impact (PI)

To estimate the price impact we use data sequences consisting of an order book snapshot, the first transaction after the snapshot and the subsequent order book snapshot. The percentage price impact is then calculated as the signed percentage change in the quote midpoint from the pre-transaction order book snapshot to the post-transaction snapshot,¹⁴ averaged over all data sequences (as defined above) in the interval

$$PI_t = \frac{1}{N_t^+ - 1} \sum_{j=1}^{N_t^+ - 1} Q_j^+ \cdot \frac{(MQ_{j+1} - MQ_j)}{MQ_j}$$

where Q_j^+ denotes the trade indicator (+1 for a buyer-initiated trade and -1 for a seller-initiated trade) of the transaction occurring after the order book snapshot j .¹⁵

- Percentage cost of a roundtrip trade ($CRT(Y)$)

To assess the liquidity for larger trades we use the order book data to calculate the weighted average prices at which a buy and a sell order of

¹³When there is more than one transaction between two order book snapshots we only use the first of these transactions.

¹⁴From the numbers in Table 1 it follows that we observe an order book snapshot every nine seconds on average. Thus, the horizon over which we calculate the price impact is slightly less than nine seconds on average. This is in line with Conrad and Wahal (2020) who recommend to use a horizon of no more than 15 seconds for liquid stocks.

¹⁵As before, when there is more than one transaction between two order book snapshots we only use the first of these transactions. We lose one observation in each interval because the last order book snapshot in an interval is discarded as it is not followed by another snapshot in the same interval.

a given size Y would execute. The weighted average price for executing a transaction of size Y USD given the current state of the order book is defined as $\frac{\sum_{k=1}^K A_k \cdot V_k}{\sum_{k=1}^K V_k}$ subject to $\sum_{k=1}^K A_k \cdot V_k = Y$ where A_k and V_k are the price and volume of the k^{th} order, respectively. Note that the K^{th} order may be subject to partial execution, depending on the outstanding dollar volume required to entirely fill the transaction volume Y . We set Y equal to the 99% quantile of the corresponding (aggregate) trade size distribution. For the currency pair BTCUSD this value is approximately equal to USD 32,100, while for ETHUSD Y roughly corresponds to USD 17,400.

To estimate the cost of a roundtrip trade of size Y , $CRT(Y)$, we calculate the weighted average prices for a market buy order and a market sell order of size Y and then express the difference between the two prices as a fraction of the quote midpoint. Finally, we calculate an equally-weighted average across all order book snapshot in interval t .

The $CRT(Y)$ measure is conceptually similar to the quoted bid-ask spread. However, while the quoted spread measures the transaction costs of a small trade (defined as a trade the size of which does not exceed the quoted depth), the $CRT(Y)$ measure estimates the execution costs of a trade of size Y .

Table 3 shows descriptive statistics for the benchmark measures for BT-CUSD obtained from daily data (results at the hourly and 15-day frequency are virtually identical and available upon request). Overall, the percentage trading costs in the cryptocurrency markets are very low. Average quoted and effective spreads on Bitfinex and Coinbase Pro are below 1 bp while the cost of a

roundtrip trade of size USD 32,100 on these two venues are below 4 bps.¹⁶ The price impact on these two exchanges amounts to approximately 40% of the effective spread, implying that the suppliers of liquidity earn a very small positive realized spread on average. Quoted spreads, effective spreads and the cost of a roundtrip trade are much larger and more volatile on Bitstamp than on the other two exchanges.¹⁷ The price impact, on the other hand, is not larger on Bitstamp than on the other two venues. Thus, suppliers of liquidity on Bitstamp appear to be earning significant realized spreads.

INSERT TABLE 3 ABOUT HERE

We also calculated correlations between our benchmark liquidity measures. Using the daily data set for the pair BTCUSD (results for other frequencies as well as for ETHUSD are similar and available upon request) we find the highest correlation between QS and ES (exchange average: 0.92) while PI has the lowest correlations with all other benchmark measures (e.g. average across exchanges between PI and QS: 0.55.). This confirms that PI captures a different dimension of market liquidity than QS and ES. Concerning the correlations between our spread measures QS and ES, respectively, and our hypothetical

¹⁶By way of comparison: Mancini et al. (2013) report liquidity for the 9 most traded exchange rates on the EBS platform over the period January 2007 to December 2009. They find EURUSD to be the most liquid rate with a mean relative quoted spread (effective spread) of 1.05 (0.31) basis points. USDCAD is the least liquid of the analyzed pairs with respect to the quoted spread (8.27 basis points) while AUDUSD has the highest effective spread (1.38 basis points).

¹⁷Because of the higher execution costs on Bitstamp traders may want to avoid Bitstamp. However, there are several reasons why we may still observe significant trading activity on Bitstamp. First, most transactions are small. The median trade size on Bitstamp is 354 USD (the corresponding values for Bitfinex and Coinbase Pro are 500 and 140 USD, respectively). Assuming a quoted spread of 6.6 bp (the median quoted spread on Bitstamp), the execution costs of a median-sized trade on Bitstamp amount to 0.14 USD (354 USD multiplied by the half-spread), an amount which traders may deem negligible. Second, there are frictions beyond the bid-ask spread. For example, trading venues differ in the ways how traders can transfer and withdraw fiat money to and from their accounts. These differences can result in cost and speed differences between the exchanges. Third, not all traders are free to choose where to trade. For example, Bitfinex did not accept US residents as customers during our sample period. Fourth, traders may prefer to trade on a venue in or close to their home country, e.g. because they are more familiar with the legislative regime.

cost of a roundtrip (CRT), we find exchange average values of 0.77 (QS with
 335 CRT) and 0.76 (ES with CRT). Note that QS is a special case of CRT measuring
 the cost of a roundtrip for trades not exceeding the quoted inside depth. The
 high correlations between QS and CRT are therefore not surprising and in line
 with similar results from stock markets (Irvine et al. (2000)). We take these
 results as evidence that QS and ES are good indicators for market liquidity not
 340 only at but also beyond the inside spread.

Figure 1 shows the evolution of the hourly benchmark measures for the
 pair BTCUSD over time. Overall the patterns reveal strong similarities in the
 liquidity measures, both within and between the three exchanges. Starting from
 higher levels in December 2017, liquidity measures continuously decrease until
 345 the end of November 2018 where they increase sharply. From this point on, the
 figure displays a significant and mostly sustainable reduction in liquidity on the
 three markets until the end of the investigation period.

INSERT FIGURE 1 ABOUT HERE

Corresponding to Table 3, Table 9 in the appendix provides descriptive
 350 statistics for the benchmark liquidity measures for the currency pair ETHUSD
 at the daily frequency (again, results for the other frequencies are essentially
 identical and are available upon request). As for BTCUSD we find the levels of
 our four benchmark measures to be very similar on Bitfinex and Coinbase Pro.
 Average quoted spreads are about twice as high as those for the pair BTCUSD.
 355 Average effective spreads for ETHUSD are below 1 bp on Bitfinex and Coinbase
 Pro, but again are higher than those for BTCUSD. Average price impacts on
 both trading venues are almost equal to effective half-spreads, implying that the
 suppliers of liquidity earn a very small realized spread on average.

As for BTCUSD, Bitstamp is substantially less liquid for ETHUSD than the

other two exchanges.¹⁸ The average quoted spread (effective spread) amounts to 13.26 bp (1.77 bp). Again the average price impact is not larger on Bitstamp than on Bitfinex and Coinbase Pro, implying substantial realized spreads to be earned by liquidity suppliers on Bitstamp.

2.2.2. Transactions-based proxy measures

As noted previously, all transactions-based liquidity measures are calculated from data on open, high, low and closing prices as well as the number of transactions and the dollar trading volume for each subinterval i . The data for the subintervals are then aggregated to one liquidity estimate for each interval t . We use the following transactions-based measures.

- Number of transactions (TX)

For each interval t we calculate the unweighted average of the number of transactions in the subintervals i , $TX_t = \frac{1}{I} \sum_i TX_{t,i}$, where I denotes the number of subintervals in interval t .

- Dollar Volume ($\$Vol$)

Our second transactions-based measure is the unweighted average of the reported dollar transaction volume $\$Vol_t = \frac{1}{I} \sum_i \$Vol_{t,i}$ in all subintervals i belonging to interval t .

- The Amihud (2002) illiquidity ratio ($Amihud$)

The Amihud (2002) illiquidity ratio for each subinterval is the absolute return (measured from the opening price to the closing price of the subinterval) divided by the dollar trading volume in the subinterval, $Amihud_t =$

¹⁸Evidently, ETHUSD is a rather infrequently traded pair on Bitstamp (roughly 5 million transactions over our investigation period, compared to roughly 24 million on Bitfinex and Coinbase Pro) which is why we only have 4,410 observations of hourly data on Bitstamp that match the 80% data availability criterion (compared to more than 11,920 one hour intervals that match this criterion on Bitstamp for the pair BTCUSD)

$\frac{1}{I} \sum_i \frac{|C_{t,i}/O_{t,i}-1|}{\$Vol_{t,i}}$, where $O_{t,i}$ and $C_{t,i}$ denote the opening and closing price in subinterval i in t , respectively. The illiquidity ratio for interval t is the unweighted average of the ratios for the subintervals i in t . We note that conceptually the illiquidity ratio is a measure of price impact. However, in empirical applications it is routinely used as a proxy for liquidity at large.

- The Roll (1984) serial covariance estimator (*Roll*)

The Roll (1984) estimator is based on the serial covariance of successive price changes. For each interval t we obtain one spread estimate from the closing prices of all subintervals i in t . If the serial covariance is positive we set the estimator to 0. We calculate two versions of Roll's measure, one based on price changes and one based on returns. The formal expression for the return-based estimator is

$$Roll_t = 2 \cdot \sqrt{-\min(cov[\frac{\Delta C_{t,i}}{C_{t,i-1}}, \frac{\Delta C_{t,i-1}}{C_{t,i-2}}], 0)}$$

where Δ is the first difference operator. In the results section, Roll_p (Roll_r) refers to the price- (return-)based version, respectively.

- The Kyle and Obizhaeva (2016) estimator (*Kyle*)

Kyle and Obizhaeva (2016) derive an illiquidity index based on the ratio of volatility to dollar volume of an asset within a given interval. It is defined as

$$Kyle_t = \left[\frac{\overline{\sigma_{t,i}^2(r)}}{\sum_i \$Vol_{t,i}} \right]^{1/3}$$

where the volatility estimator $\overline{\sigma_{t,i}^2(r)}$ is the mean of the squared returns of all subintervals i in interval t .

- The Corwin and Schultz (2012) estimator (*CS*).

The CS estimator is calculated from the high and low prices of two adjacent subintervals $i, i+1$. It is defined as

$$CS_{i,i+1} = \frac{2(\exp(\alpha)-1)}{1+\exp(\alpha)}$$

$$\alpha = \frac{\sqrt{2\beta}-\sqrt{\beta}}{3-2\sqrt{2}} - \sqrt{\frac{\gamma}{3-2\sqrt{2}}}, \beta = \left[\ln\left(\frac{H_i}{L_i}\right) \right]^2 + \left[\ln\left(\frac{H_{i+1}}{L_{i+1}}\right) \right]^2, \gamma = \left[\ln\left(\frac{H_{i,i+1}}{L_{i,i+1}}\right) \right]^2$$

410 H_i and L_i denote the high and low prices, respectively, in subinterval i , while $H_{i,i+1}$ and $L_{i,i+1}$ refer to the high and low price, respectively, of two adjacent subintervals i and $i+1$. We follow Corwin and Schultz (2012) and set negative values of the proxy to zero. The CS_t estimator for period t is the unweighted average of all CS estimators for adjacent subintervals

415 in t .

Corwin and Schultz (2012) propose a method to adjust their estimator for the overnight trading halt. We do not need to implement this modification because cryptocurrency exchanges operate 24 hours a day and seven days a week. There are thus no regular trading halts.

- 420 • The Abdi and Rinaldo (2017) estimator (AR)

Abdi and Rinaldo (2017) propose an estimator based on the natural logarithms of high, low and closing prices in subinterval i , denoted $h_i = \ln(H_i)$, $l_i = \ln(L_i)$ and $c_i = \ln(C_i)$, respectively. Further, denote by $\bar{p}_i = (h_i + l_i)/2$ the midpoint between the high and the low log prices

425 in subinterval i . We use the 'two-day corrected' version of the estimator which uses high and low price data from two adjacent subintervals i and $i+1$. It is defined as

$$AR_i = \sqrt{\max\{4(c_i - \bar{p}_i)(c_i - \bar{p}_{i+1}), 0\}}$$

The AR_t estimator for interval t is the average of the $AR_{t,i}$ measures for

430 all adjacent subintervals i in t ,

$$AR_t = \frac{1}{I-1} \sum_{i=1}^{I-1} AR_{t,i}.$$

Table 4 reports descriptive statistics for our proxy liquidity measures at the

daily frequency (results at the hourly and 15-day frequency are again available upon request). When interpreting the numbers it should be kept in mind that several measures (Roll, Kyle, Abdi and Ranaldo, and Corwin and Schultz) estimate the effective bid-ask spread while the Amihud illiquidity ratio is a measure of price impact and the two volume metrics measure the number of trades and the dollar trading volume, respectively. However, even among those measures that estimate the effective spread there are large differences. The Roll estimator delivers the largest and the Kyle and Obizhaeva (2016) estimator delivers the smallest spread estimates. We will compare the mean values shown in Table 4 to the effective spread calculated from high-frequency quote data in section 3.5 below.

INSERT TABLE 4 ABOUT HERE

For the currency pair ETHUSD descriptive statistics for our proxy liquidity measures are reported in Table 10 in the appendix. As for our benchmark measures the results indicate that the pair ETHUSD is less liquid than BTCUSD: volume-based proxy measures show lower values, while price-based measures are higher.¹⁹ We will discuss these results in more detail in section 3.7.

3. Results

We present the results in six steps. We first report time-series correlations between the transactions-based proxies and the benchmark measures. Correlations are a global measure of linear dependence. To analyze whether the dependence structure is different in the tails of the distribution we analyze, in step 2, quantile dependencies based on the empirical distribution functions. In a third

¹⁹We note that the price-based Roll measure delivers an estimate of the dollar spread, not of the percentage spread. The numerical values are lower for ETHUSD than for BTCUSD because the dollar price of ethereum is only a fraction of the bitcoin price.

step we aggregate the transactions-based measures into a composite measure, the first principal component of the transactions-based measures, and analyze whether it has higher time-series correlation with the benchmark measures than the best of the individual transactions-based measures. In order to investigate
 460 whether the performance of the transactions-based measures depends on the specific market environment we then, in step 4, split our sample along several dimensions. Specifically, we estimate time-series correlations between the proxies and the benchmark measures for the first and the second half of our sample period, for high and low return periods, high and low volatility periods
 465 and high and low volume periods. Next we report the mean absolute errors and root mean squared errors of the transactions-based measures. Finally, to capture the cross-sectional dimension, we analyze how frequently the liquidity ranking across the exchanges produced by the transactions-based measures is equal to the ranking produced by the benchmark measures.

470 In sections 3.1 to 3.6 we present results for the currency pair BTCUSD in detail; qualitative results for the pair ETHUSD are similar in most respects and are summarized in section 3.7.

3.1. Time-Series Correlations

An accurate transactions-based measure should capture the time-series variation in liquidity and should thus be positively correlated with the benchmark measures. We therefore estimate time-series correlations between the low-frequency measures and the high-frequency measures. We do so separately for three exchanges (Bitfinex, Bitstamp and Coinbase Pro) and three time frames (1-minute data (1-hour-data, daily data) aggregated to the hourly (daily, 15-
 480 daily) frequency). As it turns out, the results for the three exchanges are very

similar. We therefore report averages across the trading venues.²⁰ In the description of the results we emphasize the findings for the daily data (i.e. hourly data aggregated to the daily frequency). We believe that most researchers using transactions-based liquidity proxies will do so to obtain daily estimates, and the
 485 hourly raw data required to calculate these daily estimates are easily and freely available, e.g. from `cryptodatadownload.com`.

The results are presented in Figures 2, 3 and 4. Focusing on the results for the daily intervals (Figure 2) we find that the Abdi and Rinaldo (2017) and Corwin and Schultz (2012) estimators mostly perform best. The Corwin
 490 and Schultz (2012) estimator exhibits the highest time series correlation and the Abdi and Rinaldo (2017) estimator the second-highest correlation with the quoted spread, the price impact and the cost of a roundtrip trade, with correlations ranging from 0.48 to 0.58. The correlation with the effective spread is markedly higher, at 0.76 (0.74) for the Corwin and Schultz (2012) (Abdi and
 495 Rinaldo (2017)) estimator.

Interestingly, the number of transactions and the dollar trading volume are highly correlated with the four benchmark measures, and particularly so with the effective spread (with correlation coefficients of 0.77 and 0.78 for the number
 500 of transactions and the dollar volume, respectively, even surpassing the correlations achieved by the Corwin and Schultz (2012) and Abdi and Rinaldo (2017) estimators). What is most surprising is the sign of the coefficients. Both measures are *positively* related to the benchmark measures, implying that higher trading activity is associated with higher execution costs.

The other transactions-based proxies achieve much lower correlations. The
 505 measure that performs worst in our horse race is the Amihud (2002) illiquidity ratio. It is virtually uncorrelated with the benchmark measures, and the sign

²⁰Results for individual exchanges are available upon request.

of the correlation is even negative.²¹ The poor performance of the Amihud (2002) illiquidity ratio deserves discussion because this ratio is widely used as a measure of liquidity in empirical microstructure research. We argue that the lack of correlation between the illiquidity ratio and the benchmark measures is caused by the strong and positive relation between liquidity and trading activity discussed above. The illiquidity ratio is based on the presumption that, in a less liquid market, a given dollar trading volume will have a larger impact on prices and will thus result in a larger price change. Put differently, for a given price change higher volume points to a more liquid market and should thus be associated with lower execution costs according to the inherent logic of the measure. However, in the markets under investigation volume is *positively* related to execution costs, a relation that runs counter the logic of the illiquidity ratio. We wish to reemphasize that the finding of a positive relation between trading activity and execution costs, even though at odds with the predictions of standard theory, is not confined to the cryptocurrency markets under consideration here. Bogousslavsky and Collin-Dufresne (2020) have recently documented a similar finding for large US stocks.

INSERT FIGURE 2 ABOUT HERE

The results for the two alternative time frames (one-minute data aggregated to the hourly frequency (Figure 3) and daily data aggregated to the 15-day frequency (Figure 4)) are similar. The Corwin and Schultz (2012) estimator yields higher correlation with the benchmark measures than the other transactions-based estimators, with the Abdi and Ranaldo (2017) being a close second (and, at the hourly frequency, even ranking first for two of the benchmark measures).

²¹In their study on the foreign exchange market Karnaukh et al. (2015) find that the Amihud (2002) illiquidity ratio performs reasonably well. When calculating the illiquidity ratio the authors use the number of transactions as a proxy for the dollar trading volume because they do not have access to volume data.

The strong and positive correlations documented above for the daily data frequency between the transaction frequency and dollar trading volume on the one hand and the benchmark measures on the other hand persist at the other data frequencies.

535 The performance of the Kyle and Obizhaeva (2016) estimator is better at higher data frequencies while the Roll (1984) estimator appears to perform better at lower frequencies. The Amihud (2002) illiquidity ratio continues to be the worst-performing measure. As explained above, the most likely reason is the positive relation between trading activity and spreads in the cryptocurrency
540 markets.

INSERT FIGURE 3 ABOUT HERE

INSERT FIGURE 4 ABOUT HERE

So far we have documented differences in correlations across the different transactions-based measures, but we do not know whether the differences are
545 significant. We therefore now perform a formal test based on the Fisher r-to-z transformation. Specifically we test, separately for each of the four benchmark measures and the three trading venues, whether the correlation between the best-performing proxy and the benchmark measure is significantly higher than the correlation between the second-best performing proxy and the benchmark.
550 The results are reported in Table 5. The evidence in favor of significant differences is limited to the highest data frequency and to two benchmark measures, the quoted and the effective spread. In four out of six cases the Corwin and Schultz (2012) estimator performs best and displays significantly higher correlation with the respective benchmark measure than the second-best transactions-based measure (which, in all four cases, is the Abdi and Ranaldo (2017) estimator). In one case (effective spreads on Bitstamp) the ranking of the two
555 best-performing proxies is the same but the difference is insignificant, and in

one case (quoted spreads on Bitstamp) the order of the CS and AR estimators is reversed. For the other benchmark measures and for lower data frequencies
 560 there is no compelling evidence in favor of significant differences between the two best-performing transactions-based measures.

INSERT TABLE 5 ABOUT HERE

3.2. Quantile dependence

The correlation between the time series of transactions-based proxies and
 565 the benchmark measures of liquidity provides a global measure of dependence. However, it is conceivable that a proxy measure that fits the benchmark well in times of high liquidity (i.e. in times of low bid-ask spreads) performs poorly in times of low liquidity and vice versa. We therefore use quantile dependence to analyze the dependence structure between the benchmark and proxy measures in
 570 more detail. The quantile dependence of order q between two random variables η_p and η_b is generally defined as the conditional probability that $F_p(\eta_p)$ is smaller (greater) than q given that $F_b(\eta_b)$ is smaller (greater) than q for $q \leq 0.5$ ($q > 0.5$):²²

$$\lambda_q^{p,b} = \begin{cases} P[F_p(\eta_p) \leq q \mid F_b(\eta_b) \leq q], & \text{for } q \in (0, 0.5] \\ P[F_p(\eta_p) > q \mid F_b(\eta_b) > q], & \text{for } q \in (0.5, 1). \end{cases}$$

In our application the subscripts b and p refer to the benchmark measures
 575 and the transactions-based proxies, respectively. q denotes a quantile and F denotes the cumulative distribution function (CDF). An empirical estimate of $\lambda_q^{p,b}$ is given by

²²See e.g. Duan et al. (2019).

$$\hat{\lambda}_q^{p,b} = \begin{cases} \frac{1}{Tq} \sum_{t=1}^T \mathbf{1}_{[\hat{F}_p(\hat{\eta}_{p,t}) \leq q, \hat{F}_b(\hat{\eta}_{b,t}) \leq q]}, & \text{for } q \in (0, 0.5], \\ \frac{1}{T(1-q)} \sum_{t=1}^T \mathbf{1}_{[\hat{F}_p(\hat{\eta}_{p,t}) \leq q, \hat{F}_b(\hat{\eta}_{b,t}) \leq q]}, & \text{for } q \in (0.5, 1). \end{cases}$$

$\hat{F}_j(\hat{\eta}_j); j \in \{b, p\}$ denotes the empirical distribution functions of the benchmark and proxy measure, respectively. We estimate it using scaled ranks, i.e. we transform the data into ranks and then rescale these ranks onto the unit interval.

Intuitively, quantile dependence works as follows. For any $q \leq 0.5$ consider the $q \cdot T$ smallest observations for the benchmark measure, where T is the total number of observations. Then consider the $q \cdot T$ smallest values for a transactions-based proxy and determine the fraction of coinciding values. This fraction is the estimate of the quantile dependence, $\hat{\lambda}_q^{p,b}$.

We use data at the daily frequency²³ to estimate the quantile dependence separately for each trading venue and then calculate averages across venues. We present the results using quantile dependence plots which show the quantile dependence as a function of q . Higher quantile dependence implies a closer relation between the benchmark measures and the transactions-based proxies. The results for our four benchmark measures and eight proxies are shown in Figure 5. The dependence between the benchmark and proxy measures is generally stronger in the center of the distribution and weaker in the tails. The dependence in the tails appears to be asymmetric, it tends to be higher for larger than for smaller values.²⁴

INSERT FIGURE 5 ABOUT HERE

²³Results for the hourly frequency are qualitatively similar and are available upon request. The number of observations at the 15-day frequency is too low to reliably estimate quantile dependence, particularly in the tails of the distributions.

²⁴When interpreting the results note that a quantile dependence of 0.5 for $q = 0.5$ is expected when the two distributions are independent.

With respect to the ranking of the transactions-based liquidity measures the results from the quantile dependence analysis are consistent with the results shown in Figure 2 above. In particular, the Abdi and Rinaldo (2017) and Corwin and Schultz (2012) estimators perform very well over the entire distribution, i.e. for high as well as low levels of liquidity. The two measures of trading activity, the number of transactions and the dollar volume, perform very well when the effective spread is used as benchmark. As before, the Amihud (2002) illiquidity ratio performs poorly.

3.3. Composite Estimator

The different transactions-based measures capture different aspects of liquidity. It is, therefore, conceivable that a combination of these measures better captures the time-series variation of liquidity. To test whether this is the case we construct a composite estimator based on the eight low-frequency measures. We first standardize all variables by subtracting the mean and dividing by their standard deviation. We then extract the first principal component of the standardized data and estimate the time-series correlations between the first principal component and the benchmark measures. Table 6 shows the results for each time frame and each of the three exchanges.

INSERT TABLE 6 ABOUT HERE

The first principal component roughly explains 55% of the variation in the data. The time-series correlation is highest with the effective spread. The average (across the three exchanges) correlation ranges from 0.75 for the 15-day intervals to 0.79 for the 1-day intervals. The time-series correlation is markedly lower (with values ranging from 0.48 to 0.69) when one of the other benchmark measures is used.

Comparing the results in Table 6 to those in Figures 2, 3 and 4 reveals that the best performing individual estimators, the Corwin and Schultz (2012) and

625 Abdi and Ranaldo (2017) estimators, achieve higher time-series correlation than the composite estimator for each benchmark measure at the hourly frequency and for three out of four benchmark measures (the exception being the effective spread) at the daily data frequency. At the lowest data frequency, on the other hand, the composite estimator improves upon the best-performing individual
630 measure for three out of four benchmark measures. We thus conclude that the benefit of calculating all transactions-based proxies and aggregating them to a composite estimator is limited at least at higher data frequencies.

3.4. Sample splits

It may be the case that some of the transactions-based liquidity measures
635 perform better under specific circumstances, e.g. earlier or later in the sample period or at times of high or low volatility. To shed light on this issue we split our sample along several dimensions. The analysis is performed using the hourly data aggregated to the daily level.²⁵ We start by separately considering the first and the second half of the sample period, with the resulting sub-samples
640 covering 12/16/2017 00:00 UTC to 12/15/2018 24:00 UTC and 12/16/2018 00:00 UTC to 12/16/2019 00:00 UTC, respectively. Subsequently we split the sample into terciles according to the signed return (measured by the average of the one-hour returns within a daily interval), return volatility (measured by the standard deviation of the hour-by-hour returns within a one-day interval) and
645 the dollar volume. We then calculate separate time-series correlations for the first and the third tercile.

For each sample split we present results for all four benchmark measures. We first calculate time-series correlations for each trading venue and then average the correlations across venues. These average correlations are shown in Figures

²⁵Results for the hourly frequency are qualitatively similar and are available upon request. The number of observations at the 15-day frequency is too low to split the sample into terciles.

650 6 to 9. During the second half of the sample period the correlations between the transactions-based proxies and the benchmark measures are lower than those in the first half for most proxies (Figure 6). This reduction is much more pronounced for the quoted spread and the cost of a roundtrip trade than for the effective spread and the price impact. These results allow the conclusion that
 655 the performance of the transactions-based proxies is sufficiently stable over time when they are used to track the time-series variation of the effective spread.

The time series correlations between the transactions-based proxies and the benchmark measures do not differ much between high and low return periods (Figure 7). The performance tends to be better in low return periods for three
 660 of the four benchmark measures, the quoted and effective spread and the cost of a roundtrip trade.

When we consider the sub-samples split by volatility and volume (Figures 8 and 9, respectively) we find that the time series correlations are higher in high volatility and high volume periods for at least three of the four benchmark
 665 measures, the quoted and effective spread and the cost of a roundtrip trade.

What is most important, though, is that our previous results concerning the relative performance of the low-frequency liquidity proxies still hold. The Abdi and Ranaldo (2017) and Corwin and Schultz (2012) estimators have the best overall performance under almost all conditions. The Kyle and Obizhaeva
 670 (2016) estimator performs rather well in the low volume periods but is unable to track liquidity across high volatility periods.

INSERT FIGURE 6 ABOUT HERE

INSERT FIGURE 7 ABOUT HERE

INSERT FIGURE 8 ABOUT HERE

INSERT FIGURE 9 ABOUT HERE

675

3.5. Mean Absolute Errors and Root Mean Squared Errors

The previous analyses have focused on the ability of the transactions-based proxies to capture the time-series variability of the benchmark liquidity measures. An alternative question is whether the proxy measures are able to accurately estimate the level of the benchmark measures. Investors take the level of liquidity into account in their trading strategies and portfolio allocations. Further, because the magnitude of the execution costs determines whether a given price difference (e.g. for the same cryptocurrency at two different trading venues) can be profitably exploited, the level of liquidity is also related to market efficiency.

We use as performance metrics the prediction error between the liquidity benchmark and the liquidity proxy as measured by the root mean squared error (RMSE) and the mean absolute error (MAE). Table 7 reports the corresponding results for five transactions-based measures (Amihud, Roll_r, Kyle, CS and AR) with respect to the two benchmark measures *ES* and *PI*. We exclude as proxies the trading frequency, Roll's price based estimator and the dollar volume from the analysis because these measures are obviously unable to directly capture the percentage transaction costs.

INSERT TABLE 7 ABOUT HERE

When it comes to capturing the level of the effective spread the Kyle and Obizhaeva (2016) estimator performs very well, particularly when data at higher frequencies are used. It has the lowest RMSE and MAE for all data frequencies, i.e. hourly, daily, and 15-daily. Surprisingly, the Amihud (2002) illiquidity ratio comes closest to Kyle and Obizhaeva (2016). The other proxies capture the level of the percentage effective spread rather poorly.

When the price impact is used as benchmark measure, the Amihud (2002) illiquidity ratio yields the best results for all frequencies and both metrics for

the prediction errors with the exception of RMSE at the hourly frequency where the Kyle and Obizhaeva (2016) estimator performs best. As for the effective spread, the Roll (1984) estimator and the two measures based on high and low prices, the Abdi and Rinaldo (2017) and Corwin and Schultz (2012) estimators, are unable to capture the level of the benchmark measure.

One striking observation is that the levels of the Roll (1984), Abdi and Rinaldo (2017) and Corwin and Schultz (2012) estimators appear to strongly depend on the data frequency. All three measures deliver values which increase strongly as we move to lower data frequencies. This phenomenon as such is not new. Already in his original paper, Roll (1984) obtained much larger spread estimates from weekly than from daily data. Harris (1990) argued that the difference can partly be explained by a small sample bias in the estimator of the serial covariance.²⁶ The results in Table 7 suggest that the high-low spread estimators developed by Corwin and Schultz (2012) and Abdi and Rinaldo (2017) are subject to a similar bias.

3.6. Cross-Sectional Analysis

One potential application of transactions-based liquidity measures is to compare the liquidity of different trading venues. A good proxy measure should produce the same ranking of the venues as the benchmark measures. Therefore, in order to evaluate the low-frequency measures we simply analyze how frequently the liquidity ranking across trading venues produced by the transactions-based measures is equal to the ranking produced by the benchmark measures. We per-

²⁶Specifically, he showed that the expected value of the serial covariance estimator is $E(SCov) = \frac{-s^2}{4} - \frac{-\sigma^2}{n}$ where s is the spread, σ^2 is the variance of price changes and n is the number of observations. The bias in the serial covariance estimator, $-\frac{-\sigma^2}{n}$, increases with the square of the observation interval. Under ideal conditions (i.e. i.i.d. returns and continuous trading seven days a week, as is the rule in cryptocurrency markets), the variance of weekly price changes is seven times the variance of daily price changes while the number of observations is one seventh. Consequently, the bias in weekly data is 49 times the bias in daily data.

form the analysis separately for each exchange pair (Bitfinex/Bitstamp, Bitfinex/Coinbase Pro, and Bitstamp/Coinbase Pro) and for each time frame. For each interval (one hour, one day, 15 days) and each exchange pair we record the corresponding liquidity ranking based on the benchmark measures and based on the transactions-based proxies and then simply count the fraction of identical rankings. By chance, this fraction should be 50%. Therefore, we test whether the actual fractions are significantly larger than 50% using a simple binomial test. The results are presented in Table 8.

INSERT TABLE 8 ABOUT HERE

Two general patterns emerge. First, the results for the quoted spread, the effective spread and the cost of a roundtrip trade are much better than those for the price impact. Several of the transactions-based measures (in particular the Amihud (2002) illiquidity ratio, the Kyle and Obizhaeva (2016) measure, and the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) estimators) replicate the ranking of trading venues according to these three benchmark measures well, with fractions of correct rankings ranging up to 97.3%. For the price impact, on the other hand, the percentage of matching rankings is close to 50% for all transactions-based measures. These measures thus do not provide valuable information on the ranking of price impacts across different trading venues.

Second, the ability of some transactions-based measures to capture the cross-venue differences in liquidity depends on the data frequency. In particular, the higher the data frequency (and, correspondingly, the number of observations), the better the performance of the Abdi and Ranaldo (2017) estimator. The Amihud (2002) illiquidity ratio, the Corwin and Schultz (2012) and the Kyle and Obizhaeva (2016) estimators, in contrast, are more consistent. They perform

well at all data frequencies.²⁷

Overall, when considering all 36 comparisons (4 benchmark measures, 3 trading venue pairs, 3 data frequencies) of each transactions-based measure to the benchmark measures, the Corwin and Schultz (2012) estimator achieves a
 755 rate of correct rankings significantly more frequently than expected by pure chance (i.e., a rate of correct rankings significantly above 50%) in 26 cases. The corresponding numbers for the Kyle and Obizhaeva (2016) measure, Amihud (2002) illiquidity ratio and the Abdi and Rinaldo (2017) estimator are 23, 23 and 18, respectively.

760 3.7. Results for Ethereum

In this section we briefly discuss our results for the currency pair ETHUSD which are in most respects qualitatively similar to those for BTCUSD. As for BTCUSD, results for the three trading venues are very similar. We therefore report averages across the venues. All tables and figures we are referring to are
 765 in the appendix.

When considering the time-series correlations between our low-frequency liquidity measures and the high-frequency benchmark measures we generally find correlation levels that are slightly lower for ETHUSD than for BTCUSD, particularly at the lowest data frequency (see Figures 10 to 12). The Corwin
 770 and Schultz (2012) and Abdi and Rinaldo (2017) estimators yield the highest correlations for QS , ES and PI at the hourly and the daily frequency, with often almost identical correlation levels achieved by these two estimators. When the cost of a roundtrip trade $CRT(Y)$ ²⁸ is used as benchmark measure the Kyle

²⁷Note that the Amihud (2002) illiquidity ratio does not capture the liquidity differences between Bitfinex and Coinbase well. However, as documented in Table 3 above, the liquidity differences between these two exchanges are small. When liquidity differences are small, ranking venues according to their liquidity is less important.

²⁸We set the dollar trading volume Y to USD 17,400 which corresponds to the 99% quantile of the aggregate trade size distribution for the currency pair ETHUSD.

and Obizhaeva (2016) estimator (which performs rather poorly for the other
 775 benchmark measures) performs best. As for the pair BTCUSD the volume
 proxy measures are surprisingly highly positively correlated with the benchmark
 measures, particularly with the effective spread.

The quantile dependence plots for ETHUSD yield results similar to those
 for BTCUSD. The ranking of the low-frequency proxy measures mostly holds
 780 over the entire distribution, implying that the ranking of the proxies tends to
 be independent of the level of liquidity (see Figure 13). When we construct
 a composite estimator from the eight proxy measures by means of a principal
 component analysis we find the time-series correlations between the composite
 estimator and our benchmark measures to be mostly lower than the correlations
 785 between the benchmarks and the Corwin and Schultz (2012) and the Abdi and
 Ranaldo (2017) estimators, respectively (see Table 11). Only for the 15-day
 time frame does the composite estimator achieve higher correlations than the
 best-performing individual proxy for two of the benchmark measure (QS and
 ES).

790 We separately calculate the time-series correlations between the proxy liq-
 uidity measures and the benchmark measures for the first and the second half of
 the sample period, for high and low return, high and low volatility, and high and
 low volume periods for the daily time frame. Results are reported in Figures 14
 to 17. The time-series correlations in the first and the second half of the sample
 795 periods are similar. Thus, and in contrast to the results for BTCUSD, we do
 not find lower correlations in the second half of the sample period.

Correlations tend to be higher in high volume and high volatility periods for
 three of our benchmark measure (the exception being the price impact where
 correlations are roughly equal in high and low volume and volatility settings)
 800 while high and low return intervals yield similar correlations. Most importantly

we find that the rankings of the proxy measures remain largely unchanged in the sub-samples that result from our sample splits. The Corwin and Schultz (2012) and Abdi and Rinaldo (2017) estimators have the highest correlations with almost all benchmark measures under almost all conditions. When the cost of a roundtrip trade is used as the benchmark the Kyle and Obizhaeva (2016) estimator in some cases achieves better results than the Corwin and Schultz (2012) and Abdi and Rinaldo (2017) estimators.

To summarize, our findings relating to time-series correlations between proxy and benchmark measures for the pair ETHUSD are similar to the results for BTCUSD.

In a next step we investigate the ability of the low-frequency measures to capture the level of the high-frequency benchmarks. We use the same performance metrics as in section 3.5 and obtain results for the pair ETHUSD that are again qualitatively very similar to those for BTCUSD (see Table 12). The Kyle and Obizhaeva (2016) estimator achieves the lowest levels for RMSE and MAE for both benchmark measures (*ES* and *PI*) at the hourly frequency. At the daily and the 15-day frequency, and particularly when the price impact is used as benchmark, the Amihud (2002) illiquidity ratio performs best (except for *ES* at the daily frequency where the Kyle and Obizhaeva (2016) estimator achieves a lower RMSE), while Kyle and Obizhaeva (2016) ranks second. As for the currency pair BTCUSD we find the Roll (1984), the Abdi and Rinaldo (2017) and the Corwin and Schultz (2012) estimators to perform rather poorly. Again, the performance of these proxy measures gets worse the longer the time frame, probably because of the small sample bias mentioned on page 31.

Finally, we analyze the ability of the low-frequency measures to replicate the ranking produced by the benchmark measures as in section 3.6 above. Results are displayed in Table 13. Two measures stand out, the Amihud (2002)

illiquidity ratio and the Kyle and Obizhaeva (2016) estimator. The Corwin and
 Schultz (2012) estimator that performed well when applied to BTCUSD, does
 830 poorly when applied to ETHUSD.

4. Conclusion

In this paper we compare the performance of transactions-based liquidity
 measures to benchmark measures derived from high-frequency order book data.
 We use data for the two most actively traded cryptocurrencies, bitcoin and
 835 ethereum, and from three trading venues. We consider four benchmark mea-
 sures, (a) the quoted and (b) the effective spread, (c) the price impact, and
 (d) the cost of a roundtrip trade, and we consider the performance of the
 transactions-based measures across three dimensions, (i) their ability to cap-
 ture the time-series variation in liquidity, (ii) their ability to capture the level of
 840 liquidity, and (iii) their ability to capture cross-exchange differences in liquidity.

We find that no estimator performs well across all dimensions. The Corwin
 and Schultz (2012) and Abdi and Ranaldo (2017) estimators best capture the
 time series variation in liquidity. This is true overall, at different quantiles of
 the distribution, in the first and the second half of the sample period, and in
 845 sub-samples of high and low return, high and low volatility, and high and low
 volume periods. The measures that perform best in the cross-sectional analy-
 sis are the Amihud (2002) illiquidity ratio and the Kyle and Obizhaeva (2016)
 estimator because they do well at all data frequencies and for both currency
 pairs. When estimating the level of the benchmark measures against the Amihud
 850 (2002) illiquidity ratio and the Kyle and Obizhaeva (2016) estimator perform
 best while the Corwin and Schultz (2012) and Abdi and Ranaldo (2017) esti-
 mators do poorly in this respect.

Overall our results suggest that investors should use the Amihud (2002)

illiquidity ratio or the Kyle and Obizhaeva (2016) estimator to identify the most
855 liquid exchange. The same recommendation holds for investors when estimating
the level of execution costs in order to incorporate these into their trading
strategies. On the other hand, researchers looking for a measure that captures
the time-series variation of liquidity, or investors hoping to time the liquidity
of cryptocurrency markets and enter or exit when markets are liquid are best
860 served by the Corwin and Schultz (2012) or Abdi and Ranaldo (2017) liquidity
measures because these estimators best capture time-series variation in liquidity.
These differing findings suggest that the the setting is important in determining
the best liquidity proxy.

Our results can be used by researchers, investors, traders, and regulators to
865 understand liquidity levels and dynamics with relatively easy to acquire and pro-
cess aggregate price and volume data. In many applications, the transactions-
based aggregate measures perform adequately when describing high-frequency
measures derived from order book data. The use of these low-frequency mea-
sures is far less time-consuming and memory-intensive, offering a reasonable
870 compromise between accuracy and computational workload. Strategies that re-
quire more granular data such as triangular arbitrage or market-making will of
course require higher frequency measures.

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Tables

	number of transactions		number of order books	
	BTC	ETH	BTC	ETH
Bitfinex	37,148,069	23,820,982	7,271,422	7,336,639
Bitstamp	15,310,565	5,207,030	6,913,021	6,611,326
Coinbase Pro	38,241,727	24,420,844	8,186,287	8,186,287

Table 1: Number of transactions and order book snapshots for three exchanges and the two cryptocurrencies bitcoin (BTC) and ethereum (ETH), both traded against the USD. The sample period is 12/16/2017 to 12/16/2019.

	BTCUSD				ETHUSD			
	# TX [1,000]	dollar vol [mio USD]	$\sigma(r)$ [bp]	$\sigma(MQ)$ [bp]	# TX [1,000]	dollar vol [mio USD]	$\sigma(r)$ [bp]	$\sigma(MQ)$ [bp]
Bitfinex	51.15	149.87	13.44	13.51	32.74	57.46	16.88	15.87
Bitstamp	21.67	66.93	15.19	12.28	7.36	11.68	26.44	14.53
Coinbase Pro	54.05	87.73	12.78	13.23	34.61	40.17	15.22	15.27

Table 2: Descriptive statistics for transactions data and the quote midpoint for the pairs BTCUSD and ETHUSD. Number of transactions and dollar volume refer to daily averages, the standard deviation of price returns $\sigma(r)$ as well as the standard deviation of the quote midpoint returns $\sigma(MQ)$ refer to returns normalized to 60 seconds. The sample period is 12/16/2017 to 12/16/2019.

exchange		mean	std. dev.	Q1	median	Q3	num daily obs.
Bitfinex	QS	.7212	.6910	.2928	.4562	.9861	695
	ES	.3987	.5013	.1154	.2193	.4479	
	PI	.1514	.1029	.0803	.1257	.1901	
	CRT	3.833	1.927	2.351	3.649	4.677	
Bitstamp	QS	6.553	3.270	3.888	6.377	8.163	663
	ES	2.509	2.145	1.046	2.031	3.156	
	PI	.1779	.1438	.0834	.1366	.2386	
	CRT	13.23	5.765	9.601	11.88	15.70	
Coinbase Pro	QS	.6361	3.965	.0605	.1939	.6516	668
	ES	.4964	.6990	.0938	.2848	.6411	
	PI	.2063	.5854	.0787	.1404	.2415	
	CRT	3.407	4.202	2.272	2.945	3.911	

Table 3: Descriptive data for benchmark liquidity measures for the pair BTCUSD used in the empirical analysis at a daily resolution. The table reports descriptive statistics for the quoted spread (QS), the effective spread (ES), the price impact (PI) and the percentage cost of a roundtrip trade (CRT). Q1 (Q3) denotes the first (third) quartile. The unit of measurement is basis points. The sample period is 12/16/2017 to 12/16/2019.

exchange		mean	std. dev.	Q1	median	Q3	no. daily obs
Bitfinex	TX	53,175	45,859	22,622	37,453	67,338	695
	\$ Vol	155.1	186.9	38.01	83.25	190.3	
	Amihud	.0149	.1862	.0007	.0018	.0036	
	Roll_r	37.01	45.46	0	25.55	54.57	
	Roll_p	30.71	46.28	0	17.03	40.27	
	Kyle	.0765	.0235	.0599	.0730	.0898	
	CS	21.63	19.49	9.158	15.67	27.45	
	AR	23.60	21.23	10.62	17.02	28.81	
Bitstamp	TX	22,873	18,369	9,705	17,697	30,353	663
	\$ Vol	70.44	68.12	24.98	46.50	92.63	
	Amihud	.0378	.3474	.0019	.0029	.0060	
	Roll_r	37.03	45.66	0	25.69	52.68	
	Roll_p	31.14	48.65	0	16.15	40.18	
	Kyle	.0931	.0288	.0727	.0875	.1091	
	CS	24.25	18.12	12.14	19.20	29.73	
	AR	24.06	20.36	10.92	17.89	28.28	
Coinbase Pro	TX	56,928	33,687	35,507	46,277	67,235	668
	\$ Vol	91.78	96.97	30.50	61.93	115.1	
	Amihud	.0062	.0592	.0014	.0019	.0033	
	Roll_r	35.58	46.34	0	22.77	51.40	
	Roll_p	29.82	48.65	0	13.73	38.51	
	Kyle	.0846	.0258	.0671	.0805	.0978	
	CS	19.44	16.91	8.527	14.64	24.10	
	AR	22.74	20.02	10.13	16.66	26.74	

Table 4: Descriptive data for proxy liquidity measures for the pair BTCUSD used in the empirical analysis at a daily resolution. The table reports descriptive statistics for the number of transactions (TX), the dollar volume (\$ Vol, million USD), the Amihud measure (Amihud, values *1e6), Roll's returns based measure (Roll_r, basispoints), Roll's price based measure (Roll_p), the Kyle/Obizhaeva measure (Kyle, values *1e3), the Corwin/Schultz measure (CS, basispoints) and the Abdi/Rinaldo measure (AR, basispoints). The sample period is 12/16/2017 to 12/16/2019.

		QS	ES	PI	CRT
hourly	Bitfinex	CS (AR)**	CS (AR)**	Kyle (AR)	Kyle (AR)
	Bitstamp	AR (CS)**	CS (AR)	AR (TX)**	AR (CS)**
	Coinbase Pro	CS (AR)**	CS (AR)**	AR (CS)	CS (AR)
daily	Bitfinex	CS (AR)	CS (TX)	CS (AR)	CS (AR)
	Bitstamp	CS (AR)*	CS (TX)	CS (AR)	CS (AR)*
	Coinbase Pro	\$Vol (CS)	\$Vol (TX)	CS (\$Vol)	CS (AR)
15-daily	Bitfinex	TX (\$Vol)	TX (\$Vol)	AR (CS)	CS (AR)
	Bitstamp	\$Vol (TX)	\$Vol (TX)	\$Vol (CS)	TX (\$Vol)
	Coinbase Pro	Roll_p(\$Vol)	\$Vol (Roll_p)	AR (CS)	CS (AR)

Table 5: This table reports the best performing (in terms of correlation) proxy liquidity measure for each of the four benchmark liquidity measures (second best in parentheses) for the pair BTCUSD. **/* denote statistical significance in the difference between the best and second best measure's correlation coefficient using Fisher r-to-z transformation at the 1%/5% level. For example, the entry CS (AR)** for hourly data of Bitfinex in column QS indicates that CS features the highest correlation with the quoted spread, while AR is ranked second. These correlations are statistically different at the 1% level. The sample period is 12/16/2017 to 12/16/2019.

		expl. var	QS	ES	PI	CRT
hourly	Bitfinex	56.34	.7624	.8131	.4453	.5746
	Bitstamp	49.75	.7244	.8457	.5289	.7238
	Coinbase Pro	57.15	.5800	.6931	.6396	.6287
	mean	54.41	.6889	.7840	.5379	.6424
daily	Bitfinex	56.38	.8178	.8822	.5384	.6565
	Bitstamp	57.87	.7021	.8179	.7298	.7674
	Coinbase Pro	57.87	.0782	.6572	.1647	.2528
	mean	57.37	.5327	.7858	.4776	.5589
15-daily	Bitfinex	53.10	.7654	.8401	.5125	.5393
	Bitstamp	54.09	.7021	.7481	.7837	.7228
	Coinbase Pro	56.78	.4891	.6550	.6737	.5935
	mean	54.65	.6522	.7477	.6566	.6185

Table 6: This table reports mean correlations among the first principal component of proxy liquidity measures and the four benchmark measures for the pair BTCUSD. Column 'expl. var' shows the percentage of total variance explained by the first principal component. The sample period is 12/16/2017 to 12/16/2019.

	RMSE		MAE	
	ES	PI	ES	PI
hourly				
Amihud	.3360	.2773	.1251	.0223
Roll_r	.6787	.7663	.4081	.4598
Kyle	.1460	.0482	.0969	.0442
CS	.1949	.3274	.1116	.2091
AR	.3416	.4745	.2443	.3467
daily				
Amihud	.1599	.0344	.1136	.0180
Roll_r	5.750	5.844	3.610	3.647
Kyle	.1334	.0775	.0946	.0683
CS	2.696	2.819	2.064	2.160
AR	2.978	3.099	2.233	2.329
15-daily				
Amihud	.1555	.0213	.1184	.0182
Roll_r	31.38	31.47	21.79	21.83
Kyle	.1332	.0919	.1020	.0890
CS	15.65	15.77	13.69	13.79
AR	15.56	15.67	13.43	13.53

Table 7: This table reports average root mean squared errors (RMSE) and average mean absolute errors (MAE) for the three exchanges for the pair BTCUSD. The panels in the table refer to hourly, daily and 15-daily results respectively. Lines depict proxy measures whilst columns in the table refer to the benchmark liquidity measures effective spread (ES) and price impact (PI). Note that Amihud is conceptually a proxy for the price impact, while the other low-frequency measures Roll_r, Kyle, CS and AR estimate the spread. All values are multiplied by 1,000. The sample period is 12/16/2017 to 12/16/2019.

hourly	BF / BS	TX	\$Vol	Amihud	Roll_r	Roll_p	Kyle	CS	AR
	QS	5.32	24.19	91.78*	75.71*	75.65*	89.84*	92.64*	96.84*
	ES	5.85	24.59	91.38*	75.49*	75.43*	89.46*	92.57*	96.62*
	PI	47.66	49.54	53.39*	47.30	47.32	53.43*	55.40*	54.21*
	CRT	5.64	24.16	91.70*	75.54*	75.48*	89.84*	92.47*	96.60*
	BF / CB								
	QS	55.95*	77.52*	46.82	38.53	38.52	29.94	69.32*	72.35*
	ES	64.72*	73.41*	39.76	39.57	39.50	34.64	75.06*	73.87*
	PI	45.29	54.14*	57.06*	37.71	37.71	53.05*	55.57*	58.75*
	CRT	49.39	44.73	63.16*	37.08	37.10	59.59*	57.18*	58.99*
	BS / CB								
	QS	4.03	32.90	97.32*	76.70*	76.70*	86.83*	87.86*	96.28*
	ES	7.65	35.61	93.77*	75.59*	75.59*	83.85*	89.53*	95.05*
	PI	46.16	50.80	54.78*	49.63	49.61	54.40*	57.64*	55.69*
	CRT	4.07	32.92	97.28*	76.67*	76.67*	86.86*	87.87*	96.27*
daily	BF / BS								
	QS	0.30	20.36	81.00*	37.86	37.56	79.03*	75.72*	52.34
	ES	0.30	20.36	81.00*	37.86	37.56	79.03*	75.72*	52.34
	PI	42.38	51.89	49.77	36.65	37.41	48.11	55.05*	50.98
	CRT	0.30	20.36	81.00*	37.86	37.56	79.03*	75.72*	52.34
	BF / CB								
	QS	56.22*	85.16*	21.89	42.43	42.88	17.99	75.11*	61.62*
	ES	69.57*	79.91*	24.44	40.48	40.63	22.04	68.67*	57.87*
	PI	50.67	58.62*	42.73	37.93	38.23	43.93	53.37	51.27
	CRT	44.23	44.38	59.07*	42.28	42.43	57.87*	53.52	52.62
	BS / CB								
	QS	0.30	17.07	84.59*	42.90	42.45	83.08*	91.39*	61.18*
	ES	2.57	19.03	83.23*	42.15	41.69	81.12*	90.63*	61.63*
	PI	49.70	51.06	50.30	36.71	36.25	49.70	54.38	51.36
	CRT	0.30	17.07	84.59*	42.90	42.45	83.08*	91.39*	61.18*
15-daily	BF / BS								
	QS	0.00	19.57	82.61*	32.61	28.26	78.26*	58.70	58.70
	ES	0.00	19.57	82.61*	32.61	28.26	78.26*	58.70	58.70
	PI	39.13	54.35	47.83	23.91	23.91	43.48	50.00	50.00
	CRT	0.00	19.57	82.61*	32.61	28.26	78.26*	58.70	58.70
	BF / CB								
	QS	65.22	93.48*	8.70	39.13	43.48	15.22	78.26*	60.87
	ES	78.26*	84.78*	17.39	36.96	36.96	19.57	73.91*	69.57*
	PI	56.52	54.35	43.48	34.78	36.96	45.65	47.83	65.22
	CRT	41.30	43.48	54.35	41.30	41.30	56.52	50.00	41.30
	BS / CB								
	QS	0.00	13.04	86.96*	30.43	26.09	89.13*	84.78*	47.83
	ES	2.17	15.22	84.78*	32.61	28.26	86.96*	86.96*	50.00
	PI	54.35	54.35	45.65	28.26	32.61	47.83	47.83	41.30
	CRT	0.00	13.04	86.96*	30.43	26.09	89.13*	84.78*	47.83

Table 8: This table reports the percentage of matching orders of proxy and benchmark liquidity measures for paired exchange-wise comparisons for the pair BTCUSD. Panels depict results for hourly, daily and 15-daily intervals respectively. Each panel reports results for the three respective pairs for Bitfinex (BF), Bitstamp (BS) and Coinbase Pro (CB). For instance, 96.84 in the hourly panel, for the pair Bitfinex / Bitstamp (BF / BS), for the quoted spread (QS) and the Abdi and Rinaldo estimator (AR) implies that the Abdi and Rinaldo estimator matches the order of the contemporaneous order of the quoted spread for this exchange pair in 96.84% of all intervals. An asterisk (*) indicates that the corresponding value is significantly higher (at the 1% level) than 50% (the value that would obtain from guessing). The sample period is 12/16/2017 to 12/16/2019.

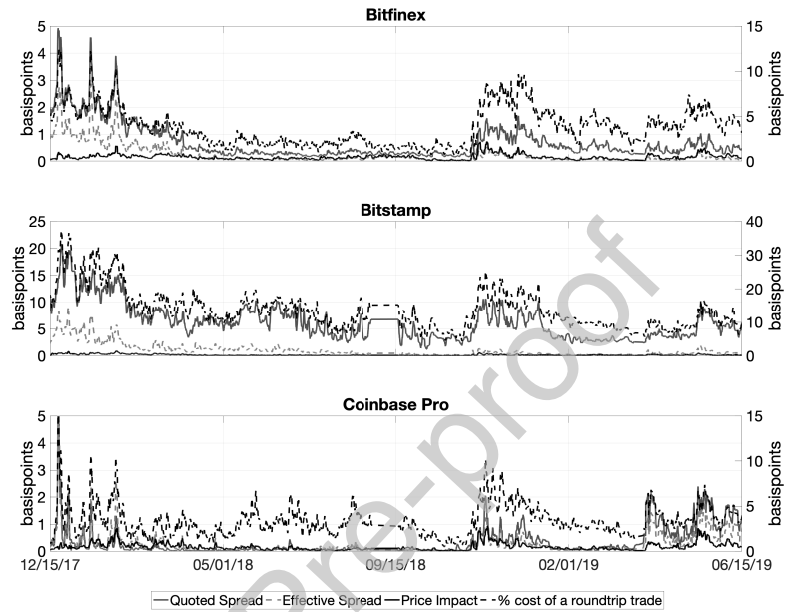
980 **Figures**

Figure 1: This figure plots a 24 hours moving average of the benchmark liquidity measures for the pair BTCUSD at an hourly resolution. Quoted spread (gray solid line), effective spread (light gray dashed line) and price impact (black solid line) are shown on the left axis, the percentage cost of a roundtrip trade (black dashed line) is shown on the right axis. The unit of measurement is basis points. The sample period is 12/16/2017 to 12/16/2019. Note that for Bitstamp and Coinbase Pro we have a 19 days lack of data in September 2018.

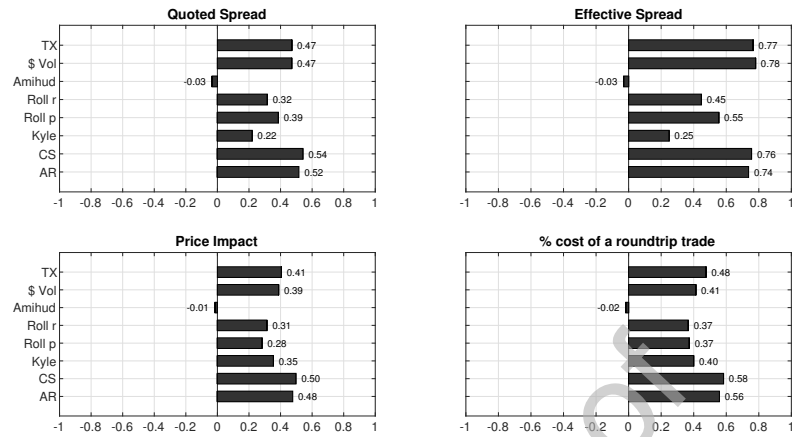


Figure 2: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

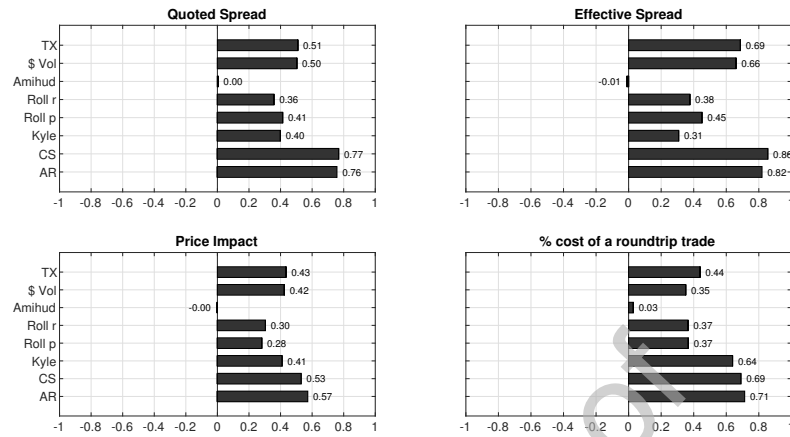


Figure 3: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on an hourly basis over the sample period 12/16/2017 to 12/16/2019.

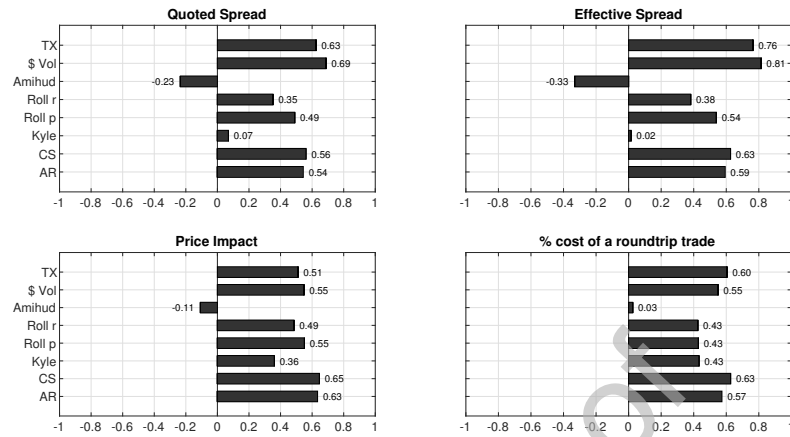


Figure 4: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a 15-daily basis over the sample period 12/16/2017 to 12/16/2019.

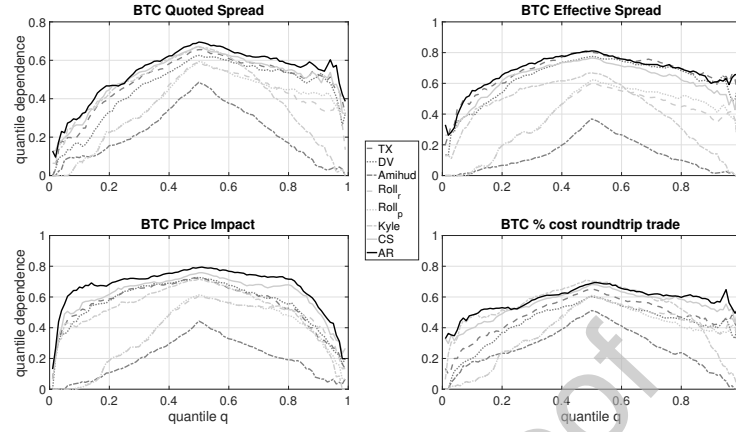


Figure 5: This figure plots the quantile dependence (averaged across the three trading venues) as a function of the quantile q in steps of 0.01 for the four benchmark liquidity measures and each of the 8 proxy measures for the pair BTCUSD for daily data. The sample period is 12/16/2017 to 12/16/2019.

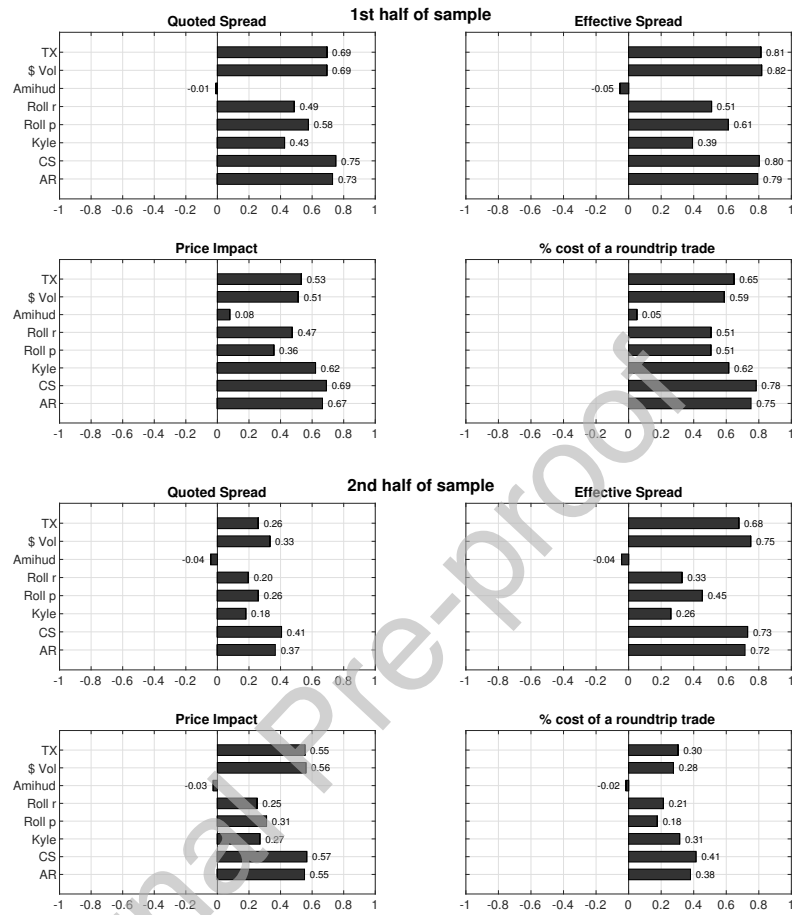


Figure 6: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the first and second half of the sample for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

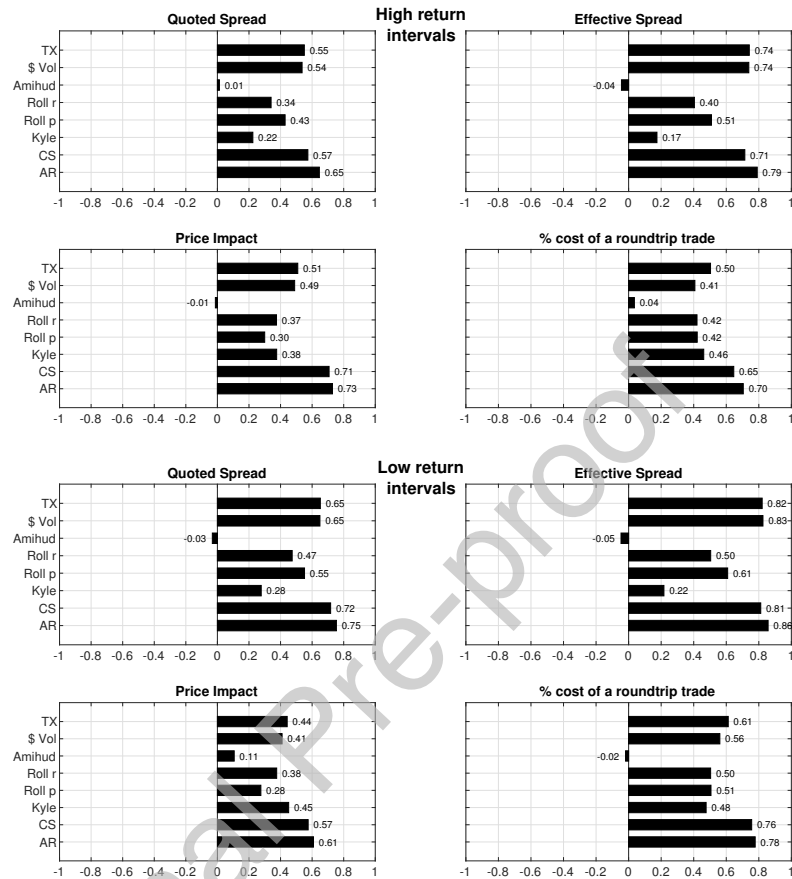


Figure 7: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures in the subset of high and low return intervals for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

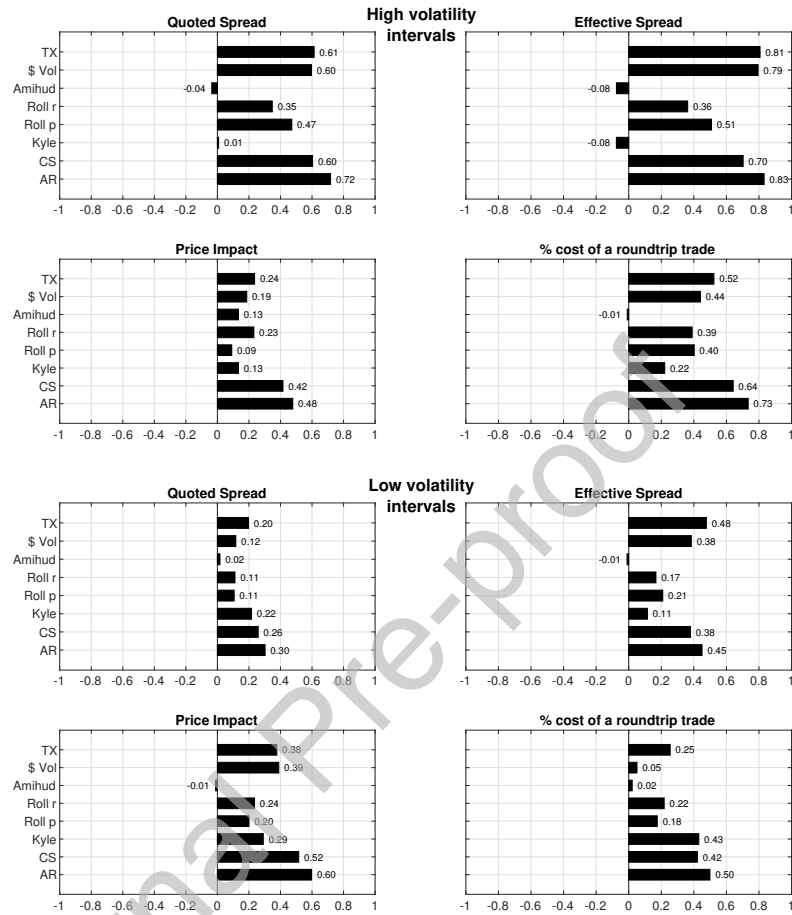


Figure 8: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures in the subset of high and low volatility intervals for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

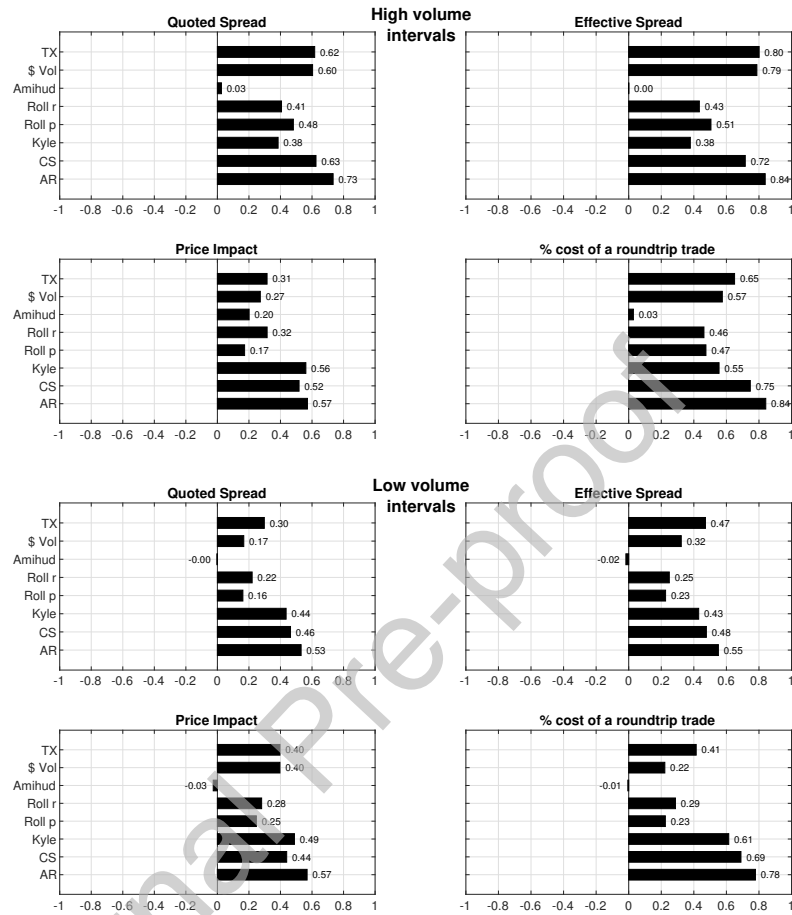


Figure 9: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures in the subset of high and low volume intervals for the pair BTCUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

Appendix

This appendix reports the results for the currency pair ETHUSD.

exchange		mean	std. dev.	Q1	median	Q3	num daily obs.
Bitfinex	QS	1.367	1.433	.6406	1.020	1.420	699
	ES	.5393	.9703	.1551	.2503	.4246	
	PI	.2467	.1735	.1364	.1983	.3066	
	CRT	7.224	3.809	3.854	7.634	9.885	
Bitstamp	QS	13.26	5.032	9.946	12.36	15.31	651
	ES	1.774	2.150	.5940	1.184	2.078	
	PI	.2618	.2635	.1032	.1971	.3517	
	CRT	30.33	9.934	23.17	29.15	35.24	
Coinbase	QS	1.232	.9283	.6471	.9791	1.475	676
	ES	.7068	.9177	.3027	.4542	.7668	
	PI	.3320	.2726	.1631	.2588	.4091	
	CRT	7.447	3.125	5.351	6.916	8.988	

Table 9: Descriptive data for benchmark liquidity measures for the pair ETHUSD used in the empirical analysis at a daily resolution. The table reports descriptive statistics for the quoted spread (QS), the effective spread (ES), the price impact (PI) and the percentage cost of a roundtrip trade (CRT). The unit of measurement is basis points. The sample period is 12/16/2017 to 12/16/2019.

exchange		mean	std. dev.	Q1	median	Q3	no. daily obs
Bitfinex	TX	33,985	25,895	13,948	27,570	47,358	699
	\$ Vol	59.24	72.87	13.88	33.20	77.84	
	Amihud	.0443	.4363	.0025	.0056	.0162	
	Roll_r	48.51	56.57	0	37.30	70.78	
	Roll_p	2.006	3.882	0	.7249	1.906	
	Kyle	.1312	.0414	.1006	.1238	.1560	
	CS	26.82	21.74	13.16	20.58	34.00	
	AR	30.32	24.20	15.60	23.83	36.51	
Bitstamp	TX	7,889	8,127	2,947	5,451	10,075	651
	\$ Vol	12.47	17.26	3.633	7.136	14.21	
	Amihud	.2857	1.157	.0189	.0407	.1101	
	Roll_r	49.01	54.73	0	36.74	69.82	
	Roll_p	2.089	3.957	0	.7469	2.046	
	Kyle	.2124	.0598	.1714	.2026	.2417	
	CS	28.46	22.14	14.63	22.35	35.32	
	AR	31.13	23.34	16.13	24.50	38.08	
Coinbase Pro	TX	35,894	31,307	14,997	26,626	46,208	676
	\$ Vol	41.60	63.87	10.11	21.31	45.18	
	Amihud	.5676	10.38	.0045	.0087	.0165	
	Roll_r	46.42	54.08	0	33.90	67.08	
	Roll_p	1.919	3.715	0	.6591	1.826	
	Kyle	.1427	.0402	.1160	.1369	.1634	
	CS	26.10	20.54	13.13	20.65	32.34	
	AR	29.80	23.62	15.07	23.43	36.01	

Table 10: Descriptive data for proxy liquidity measures for the pair ETHUSD used in the empirical analysis at a daily resolution. The table reports descriptive statistics for the number of transactions (TX), the dollar volume (\$ Vol, million USD), the Amihud measure (Amihud, values *1e6), Roll's returns based measure (Roll_r, basispoints), Roll's price based measure (Roll_p), the Kyle/Obizhaeva measure (Kyle, values *1e3), the Corwin/Schultz measure (CS, basispoints) and the Abdi/Rinaldo measure (AR, basispoints). The sample period is 12/16/2017 to 12/16/2019.

		expl. var	QS	ES	PI	CRT
hourly	Bitfinex	50.98	.6967	.7518	.4151	.4142
	Bitstamp	50.23	.6358	.8811	.5260	.5481
	Coinbase Pro	51.80	.5314	.7141	.5308	.3587
	mean	51.00	.6213	.7823	.4906	.4403
daily	Bitfinex	52.35	.6790	.7769	.4243	.2313
	Bitstamp	54.05	.6600	.8885	.7036	.5367
	Coinbase Pro	52.24	.3456	.5627	.4104	.1772
	mean	52.88	.5615	.7427	.5128	.3151
15-daily	Bitfinex	56.29	.4751	.6511	.1160	-.1912
	Bitstamp	50.21	.7151	.8200	.7034	.4935
	Coinbase Pro	54.80	-.0317	.3570	.1493	-.2930
	mean	53.76	.3862	.6094	.3229	.0031

Table 11: This table reports mean correlations among the first principal component of proxy liquidity measures and the four benchmark measures for the pair ETHUSD. Column 'expl. var' shows the percentage of total variance explained by the first principal component. The sample period is 12/16/2017 to 12/16/2019.

	RMSE		MAE	
	ES	PI	ES	PI
hourly				
Amihud	63.05	63.03	1.524	1.398
Roll_r	1.028	1.149	.6709	.7311
Kyle	.2013	.0863	.1202	.0785
CS	.2336	.3955	.1245	.2153
AR	.4731	.6465	.3472	.4729
daily				
Amihud	.1685	.0370	.1007	.0286
Roll_r	7.203	7.281	4.758	4.788
Kyle	.1563	.1414	.1177	.1343
CS	3.316	3.428	2.612	2.684
AR	3.716	3.827	2.941	3.014
15-daily				
Amihud	.1494	.0344	.1039	.0299
Roll_r	41.89	41.97	32.97	33.02
Kyle	.1634	.1765	.1434	.1704
CS	19.31	19.40	17.63	17.71
AR	19.54	19.62	17.56	17.63

Table 12: This table reports average root mean squared errors (RMSE) and average mean absolute errors (MAE) for the three exchanges for the pair ETHUSD. The panels in the table refer to hourly, daily and 15-daily results respectively. Lines depict proxy measures whilst columns in the table refer to the benchmark liquidity measures effective spread (ES) and price impact (PI). Note that Amihud is conceptually a proxy for the price impact, while the other low-frequency measures Roll_r, Kyle, CS and AR estimate the spread. All values are multiplied by 1,000. The sample period is 12/16/2017 to 12/16/2019.

hourly	BF / BS	TX	\$Vol	Amihud	Roll_r	Roll_p	Kyle	CS	AR
	QS	0.34	4.99	93.75*	68.41*	68.21*	97.02*	57.29*	84.62*
	ES	2.13	6.59	92.56*	67.68*	67.48*	95.38*	58.16*	84.57*
	PI	39.32	40.12	60.95*	48.71	48.78	60.47*	57.68*	62.35*
	CRT	0.37	4.97	93.73*	68.39*	68.18*	97.05*	57.27*	84.60*
	BF / CB								
	QS	51.51*	50.83	54.79*	40.19	40.22	54.40*	68.65*	69.07*
	ES	55.71*	45.99	49.47	39.81	39.74	57.40*	74.73*	71.66*
	PI	51.51*	45.03	56.26*	39.61	39.52	59.19*	60.74*	61.80*
	CRT	46.90	31.09	68.88*	41.19	41.20	71.84*	65.24*	66.38*
	BS / CB								
	QS	0.55	6.30	97.27*	66.39*	66.44*	96.98*	47.54	76.10*
	ES	11.32	16.30	86.90*	63.71*	63.73*	86.70*	57.87*	79.38*
	PI	54.98*	54.82*	46.02	43.52	43.50	45.45	64.73*	56.32*
	CRT	0.55	6.30	97.27*	66.39*	66.44*	96.98*	47.54	76.10*
daily	BF / BS								
	QS	0.00	0.31	97.85*	37.23	36.92	99.69*	54.00	53.08
	ES	0.15	0.46	97.69*	37.23	36.92	99.54*	54.15	53.23
	PI	56.62*	56.62*	44.62	40.00	40.31	43.38	53.08	52.46
	CRT	0.00	0.31	97.85*	37.23	36.92	99.69*	54.00	53.08
	BF / CB								
	QS	47.02	48.81	52.08	37.50	36.90	54.17	56.70*	51.49
	ES	52.38	44.35	58.04*	35.57	34.38	57.74*	54.32	49.11
	PI	54.46*	41.67	59.23*	35.12	34.97	58.63*	53.72	49.11
	CRT	38.54	27.53	73.96*	35.86	36.76	75.74*	50.00	52.23
	BS / CB								
	QS	0.00	0.31	97.69*	44.31	44.77	99.69*	58.92*	54.62*
	ES	17.69	18.00	80.62*	42.62	42.77	82.00*	69.85*	54.77*
	PI	76.92*	76.92*	24.77	37.54	36.77	22.77	51.85	50.92
	CRT	0.00	0.31	97.69*	44.31	44.77	99.69*	58.92*	54.62*
15-daily	BF / BS								
	QS	0.00	0.00	100.00*	39.13	39.13	100.00*	45.65	45.65
	ES	0.00	0.00	100.00*	39.13	39.13	100.00*	45.65	45.65
	PI	63.04	63.04	36.96	41.30	41.30	36.96	52.17	39.13
	CRT	0.00	0.00	100.00*	39.13	39.13	100.00*	45.65	45.65
	BF / CB								
	QS	43.48	50.00	50.00	43.48	32.61	54.35	43.48	50.00
	ES	45.65	47.83	52.17	41.30	32.61	56.52	32.61	52.17
	PI	52.17	41.30	58.70	43.48	39.13	58.70	30.43	41.30
	CRT	34.78	28.26	71.74*	43.48	39.13	76.09*	39.13	45.65
	BS / CB								
	QS	0.00	0.00	100.00*	30.43	34.78	100.00*	69.57*	56.52
	ES	13.04	13.04	86.96*	36.96	36.96	86.96*	73.91*	56.52
	PI	84.78*	84.78*	15.22	50.00	47.83	15.22	41.30	45.65
	CRT	0.00	0.00	100.00*	30.43	34.78	100.00*	69.57*	56.52

Table 13: This table reports the percentage of matching orders of proxy and benchmark liquidity measures for paired exchange-wise comparisons for the pair ETHUSD. Panels depict results for hourly, daily and 15-daily intervals respectively. Each panel reports results for the three respective pairs for Bitfinex (BF), Bitstamp (BS) and Coinbase Pro (CB). For instance, 84.62 in the hourly panel, for the pair Bitfinex / Bitstamp (BF / BS), for the quoted spread (QS) and the Abdi and Ranaldo estimator (AR) implies that the Abdi and Ranaldo estimator matches the order of the contemporaneous order of the quoted spread for this exchange pair in 84.62% of all intervals. An asterisk (*) indicates that the corresponding value is significantly higher (at the 1% level) than 50% (the value that would obtain from guessing). The sample period is 12/16/2017 to 12/16/2019.

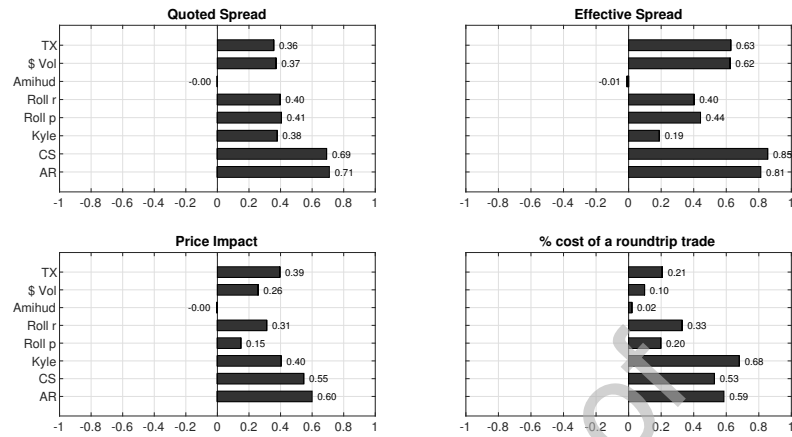


Figure 10: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on an hourly basis over the sample period 12/16/2017 to 12/16/2019.

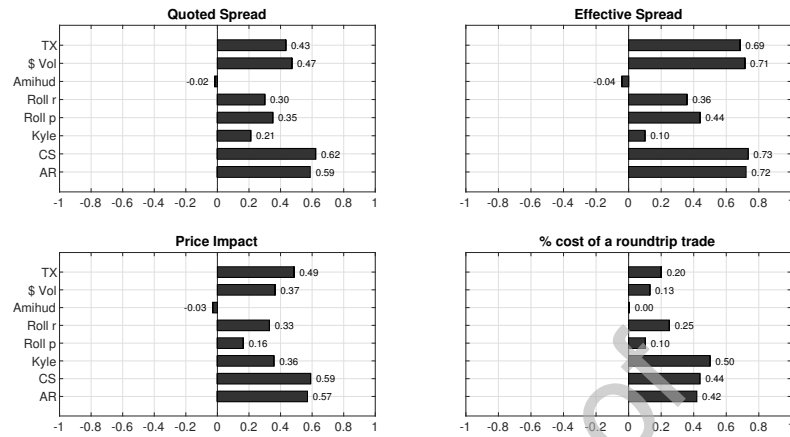


Figure 11: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

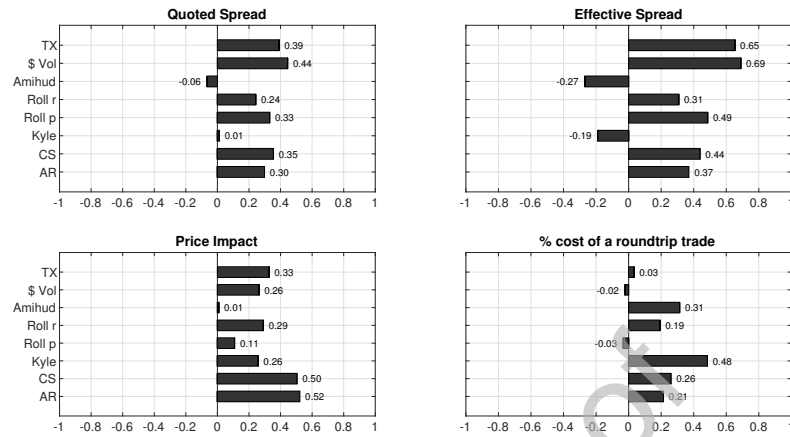


Figure 12: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a 15-daily basis over the sample period 12/16/2017 to 12/16/2019.

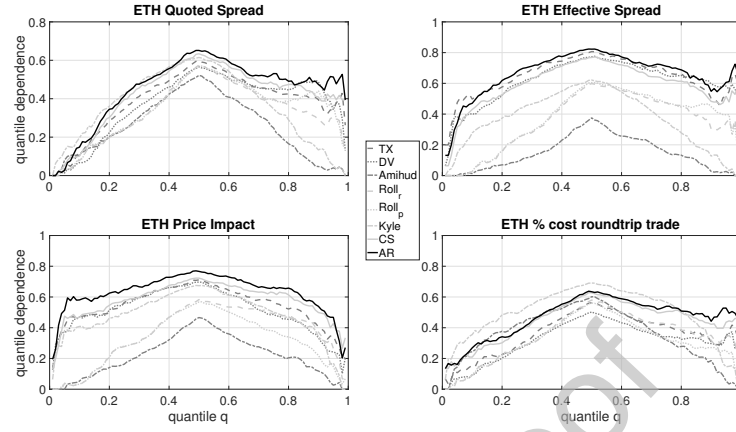


Figure 13: This figure plots the quantile dependence (averaged across the three trading venues) as a function of the quantile q in steps of 0.01 for the four benchmark liquidity measures and each of the 8 proxy measures for the pair ETHUSD for daily data. The sample period is 12/16/2017 to 12/16/2019.

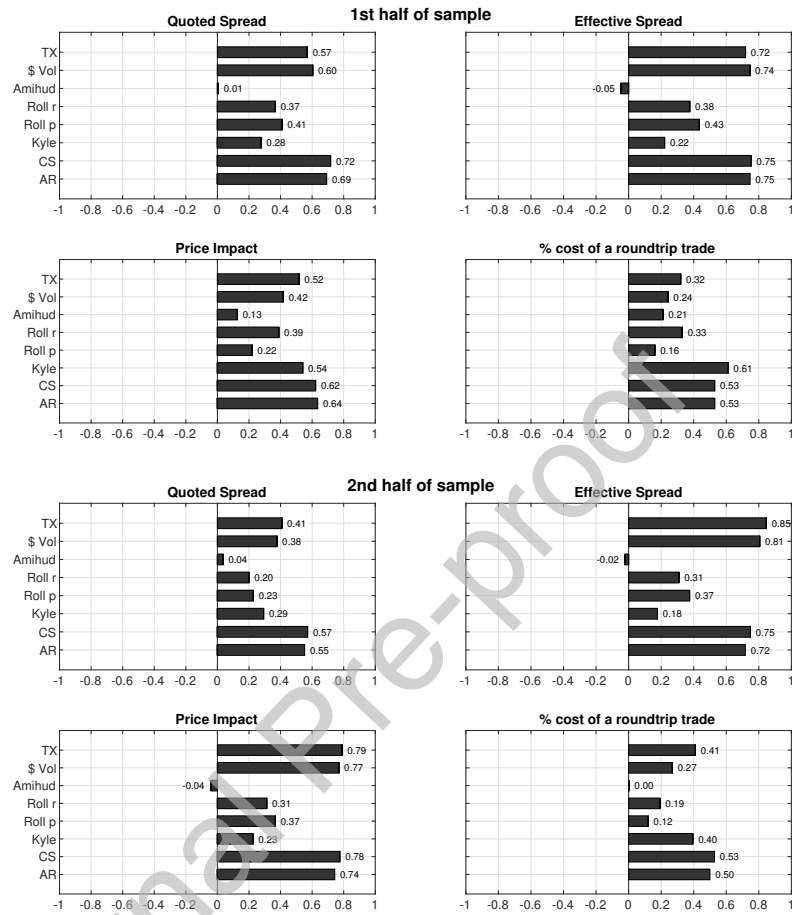


Figure 14: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures for the first and second half of the sample for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

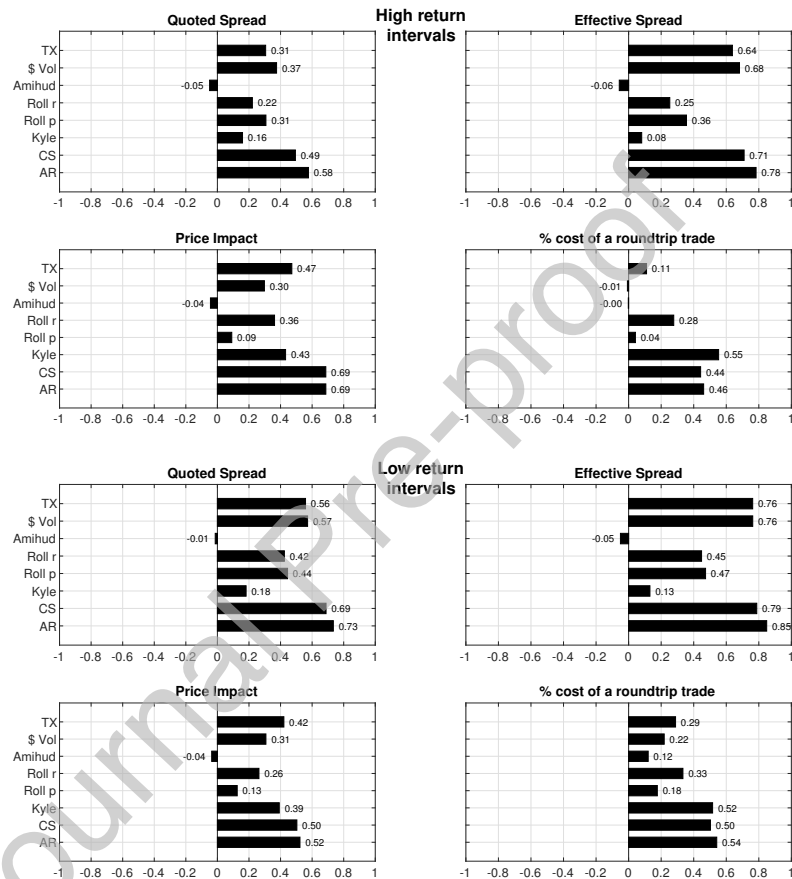


Figure 15: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures in the subset of high and low return intervals for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

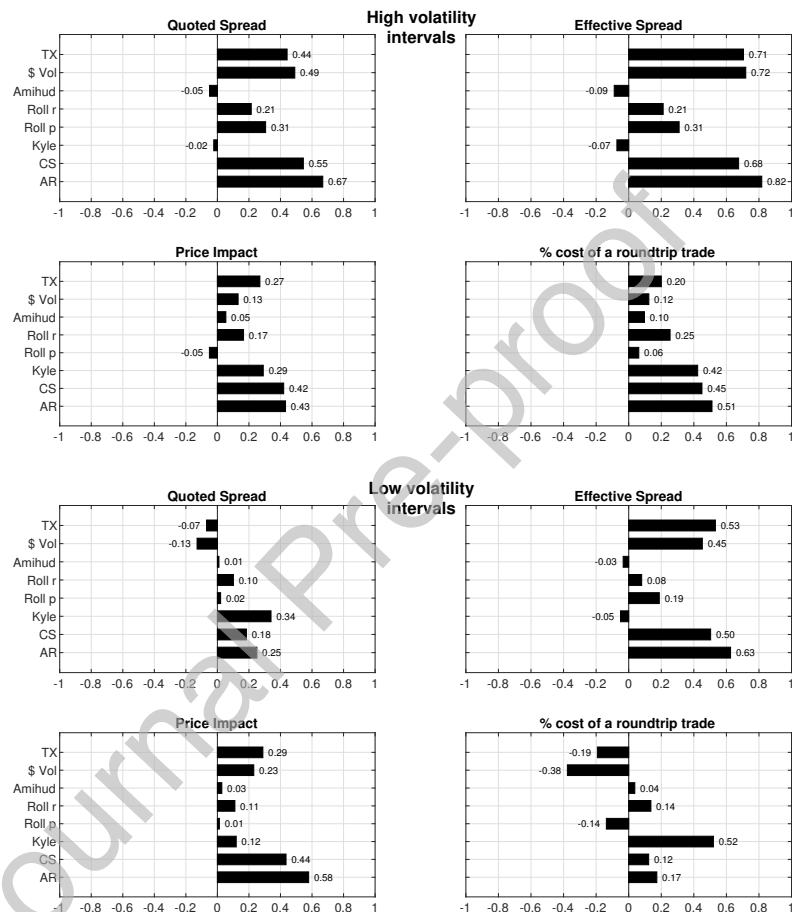


Figure 16: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures in the subset of high and low volatility intervals for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

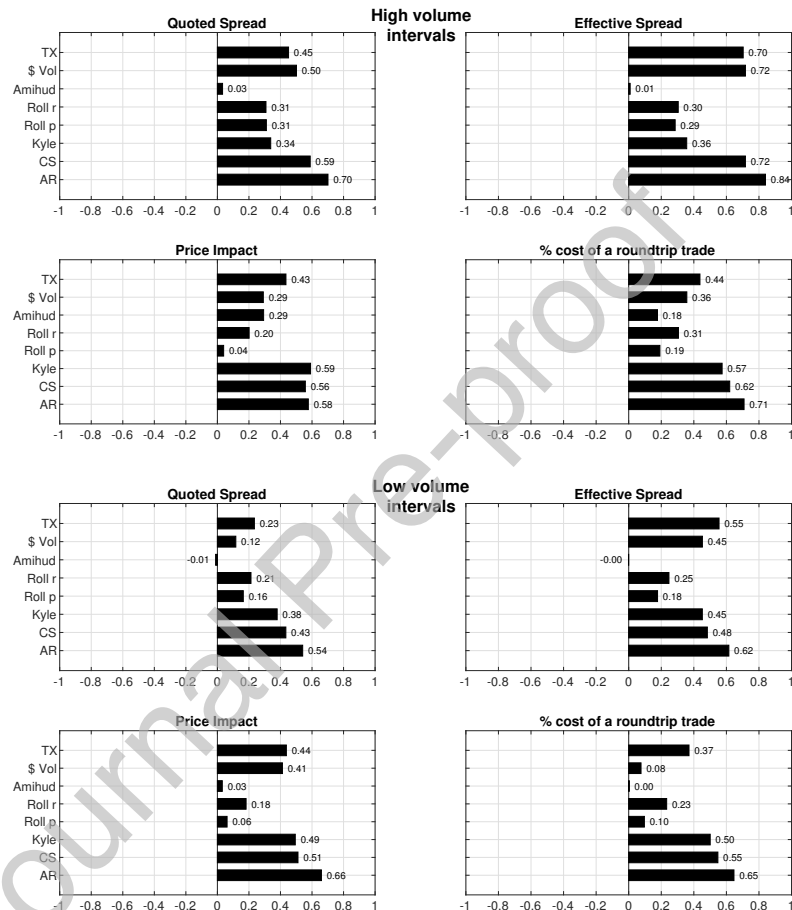


Figure 17: This figure shows time series correlations between benchmark measures of liquidity and low-frequency proxy measures in the subset of high and low volume intervals for the pair ETHUSD. Values represent simple averages across the three exchanges Bitfinex, Bitstamp, and Coinbase Pro. Liquidity measures are calculated on a daily basis over the sample period 12/16/2017 to 12/16/2019.

Author Statement

Brauneis: Methodology, Software, Formal analysis, Investigation, Data Curation, Writing, Visualization Mestel: Methodology, Formal analysis, Writing, Visualization, Project administration Riordan: Conceptualization, Methodology, Formal analysis, Writing Theissen: Conceptualization, Methodology, Formal analysis, Writing

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